

Highlights

Thermal maintenance, not hypothermia, governs mysticete cold-water thermoregulation: a comparative 3D finite-element bioheat analysis

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- Three-dimensional finite-element bioheat model of seven mysticete species with an explicit counter-current thermoregulation framework, including appendage-root retia and a peripheral counter-current shell, anchored to the stratified minke temperature transect of Folkow & Blix (1992)
- The model reproduces the measured thermal structure (deep body 35.0 °C, core/blubber interface 28.6 °C, near-water skin), supporting the insulation and vascular physics
- Under a defended arterial-set-point formulation, all species, modes and temperatures maintain physiological core temperature; cold water imposes an energetic cost, not acute modelled hypothermia
- Thermal-maintenance energy requirement is quantified per species and water temperature (from ~24 to ~1340 Mcal day⁻¹); mass-specifically the heavily blubbered balaenids require ~1/3 the energy per unit body mass of the thermal-window rorquals, and demand peaks in the mid-cold range
- The heat increment of feeding supplies a temperature-dependent share of the thermoregulatory demand, so feeding is itself thermoregulatory

Thermal maintenance, not hypothermia, governs mysticete cold-water thermoregulation: a comparative 3D finite-element bioheat analysis

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ABSTRACT

Baleen whales (Mysticeti) maintain core body temperatures of 33 °C to 37 °C while inhabiting waters as cold as −2 °C, yet the interplay between morphology, insulation, and active vascular control that enables this feat remains poorly quantified across species. We present a three-dimensional finite element model based on the Pennes bioheat equation that simulates transient thermal fields within anatomically realistic whale body geometries. Active thermoregulatory control is driven by ambient water temperature relative to each species' thermoneutral zone via a sigmoid control signal $u = \tanh[(T_w - T_{in})/\lambda]$ that continuously modulates appendage counter-current heat-exchange effectiveness and window perfusion; the water-side convection coefficient is not the thermoregulatory control variable, and core perfusion is unaffected. Blood perfusion rates are based on measured cetacean tissue values (Hokkanen, 1990): the core retains full arterial perfusion ($\omega_b = 0.8 \text{ kg m}^{-3} \text{ s}^{-1}$) while blubber perfusion is strongly reduced to represent counter-current heat exchange. Species-specific parameters are drawn from peer-reviewed literature and applied to seven mysticete species: humpback whale (*Megaptera novaeangliae*), North Atlantic right whale (*Eubalaena glacialis*), blue whale (*Balaenoptera musculus*), bowhead whale (*Balaena mysticetus*), fin whale (*Balaenoptera physalus*), gray whale (*Eschrichtius robustus*), and sei whale (*Balaenoptera borealis*). The core matrix comprises 252 simulations: nine water temperatures (−2 °C to 30 °C) $\times$ four thermoregulatory modes (active, no-control, swimming with and without active control) $\times$ seven species. It is complemented by food-availability and body-condition extensions that model reduced metabolic heat and emaciation. An eighth species, the common minke whale, is swept through the same grid but carried separately as a physiological validation case, using proxy geometry, against the only published stratified cetacean blubber-temperature transect (Folkow and Blix, 1992), bringing the main comparative and validation matrix to 793 finite-element simulations. We additionally perform a targeted 48-case humpback sensitivity audit for the weakest-constrained thermoregulatory parameters. We introduce a thermoregulatory biomarker suite including the Thermoregulatory Effort Score (TES, −1 to +1), heat-balance efficiency, thermal window fraction, the thermal-maintenance energy requirement, and a physiological thermal-limit check that reports whether the defended arterial-set-point formulation remains within physiological bounds at every simulated temperature.

A clear spectrum of strategies emerges from the corrected model. Across all seven species, four thermoregulatory modes and nine water temperatures, no simulation under the defended arterial-set-point formulation crosses a physiological temperature limit: within this model structure, mysticetes are not driven to acute hypothermia by cold water but by the *energetic cost* of staying warm. We therefore recast the output as a thermal-maintenance energy requirement. This is the metabolizable energy needed to hold the core set-point at each water temperature, reported as maintenance-equivalent heat-loss costs. Thermal-maintenance requirements range from $\sim 24 \text{ Mcal day}^{-1}$ for the minke validation species in warm water to $\sim 1340 \text{ Mcal day}^{-1}$ for the blue whale in near-freezing water, scaling with body size. On a mass-specific basis, however, the heavily blubbered balaenids (bowhead, right whale) are the most economical, requiring roughly a third of the energy per unit body mass of the thermal-window-reliant rorquals. This is the quantitative signature of the passive-insulation versus active-window strategies. For most species the requirement peaks not at the coldest temperature but in the mid-cold range, where active narrowing of the appendage thermal windows has saturated. The heat increment of feeding supplies a temperature-dependent fraction of the thermoregulatory demand, so that ingestion is itself a thermoregulatory act. A 48-case humpback sensitivity audit kept defended core temperature within 0.12 °C of the baseline while shifting maintenance-equivalent energy costs by up to 22%, confirming that the central thermal-maintenance result is robust but that absolute energy magnitudes remain parameter-sensitive. These quantitative biomarkers provide a comparative framework for assessing species-specific thermal vulnerability to ocean warming and prey decline through energy requirement rather than acute thermal failure.

1. Introduction

Marine mammals face a fundamental thermal challenge: water conducts heat approximately 25 times more efficiently than air (Schmidt-Nielsen, 1997), and baleen whales (Mysticeti) spend their entire lives submerged in or surrounded by seawater that is almost always cooler than their core body temperature. Unlike pinnipeds, which can haul out onto land or ice, mysticetes have no terrestrial refuge from convective heat loss. They have evolved a suite of morphological and physiological adaptations to manage their thermal budget, the most prominent being a thick subcutaneous blubber layer that serves simultaneously as insulation, energy reserve, and hydrodynamic streamlining (Pabst, Rommel and McLellan, 1999; Koopman, 2007). Baleen whales are particularly interesting from a thermoregulatory perspective because the species modelled here span two major extant families: Balaenopteridae (rorquals) and Balaenidae (right and bowhead whales). These families differ markedly in body plan, insulation strategy, and habitat temperature regime.

The thermal biology of individual cetacean species has been studied through post-mortem measurements (Kanwisher and Sundnes, 1966; Brodie and Paasche, 1985), infrared thermography (Cuyler, Wiulsdorf and Oritsland, 1992; Heath and Ridgway, 1999), and theoretical models (Kvadsheim, Folkow and Blix, 1997; Hokkanen, 1990; Watts, Hansen and Lavigne, 1993). However, systematic cross-species comparisons using a common modelling framework remain scarce. Most existing thermal models are one-dimensional, treating the whale body as a radially symmetric cylinder or sphere with uniform insulation (Hokkanen, 1990; Watts et al., 1993). While these models capture the essential physics of heat conduction through blubber, they cannot represent the complex three-dimensional anatomy of tapered peduncles, thin appendages, and ventral grooves. These structures create spatially heterogeneous thermal fields and enable active thermoregulation through regional blood-flow control.

The importance of appendages as “thermal windows” was recognised early by Scholander and Schevill (1955), who described counter-current heat exchangers (CCHE) in whale flippers that can recover more than 90% of arterial heat when vasoconstricted. Subsequent anatomical studies confirmed elaborate vascular networks in the flukes (Rommel, Pabst, McLellan, Mead and Potter, 1995), dorsal fins (Pabst, Rommel and McLellan, 2002), and even the tongue and palate of bowhead whales (Ford, Werth and George, 2013). Infrared thermography has shown that cetacean appendages can be several degrees warmer than the surrounding water when the animal is heat-stressed, directly visualising active heat dumping (Heath and Ridgway, 1999; Barbieri, McLellan, Wells, Blum, Hofmann, Hohn and Rowles, 2010). Despite this body of qualitative evidence, the relative contributions of insulation thickness, thermal conductivity, metabolic heat generation, and vascular modulation to the overall thermal balance have not been partitioned quantitatively across species with different body plans and ecological niches.

Climate change adds urgency to this question. Ocean warming is altering the thermal landscapes that whales have evolved to exploit (Hazen, Jorgensen, Rykaczewski, Bograd, Foley, Jonsen, Shaffer, Dunne, Costa, Crowder and Block, 2013). Species that are already near their upper thermal tolerance during warm-water breeding seasons may face compounding energetic costs. North Atlantic right whales are a clear example, because their habitat has shifted northward (Record, Runge, Pendleton, Balch, Davies, Pershing, Johnson, Stamieszkin, Ji, Feng, Kraus, Kenney, Hudak, Mayo, Chen, Salisbury and Thompson, 2019). These energetic costs arise from increased thermoregulatory demands at a time when prey availability is also declining (Meyer-Gutbrod and Greene, 2015). Bioenergetic models that underpin population viability assessments require accurate estimates of basal and thermoregulatory metabolic expenditure (Fortune, Trites, Mayo, Rosen and Hamilton, 2013; Christiansen, Vivier, Charlton, Ward, Amerson, Burnell and Bejder, 2018), yet these estimates are sensitive to assumptions about insulation effectiveness and vascular control capacity that vary markedly among species.

In this study, we develop a three-dimensional finite element thermal model based on the Pennes bioheat equation (Pennes, 1948) and apply it to seven fully simulated mysticete species spanning three families (Balaenopteridae, Balaenidae, and Eschrichtiidae), with a range of body sizes, insulation strategies, and habitat temperature regimes (Table 1), plus a common minke whale carried separately as a validation case. We parameterise each species from published morphometric, histological, and physiological data, simulate transient thermal fields across the full range

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of ecologically relevant water temperatures (-2°C to 30°C), and compare the resulting heat loss rates, core-to-skin temperature gradients, thermoregulatory effort, and food-stress vulnerability across species. Our objectives are to:

1. quantify how insulation morphology (blubber thickness, conductivity) and active vascular control jointly determine the thermal envelope of each species using a thermoregulatory biomarker framework;
2. identify which species are most sensitive to water temperature changes and assess their thermal sensitivity across the ecologically relevant range;
3. assess the relative importance of appendage-mediated heat dissipation (thermal windows) across species with and without dorsal fins and ventral grooves;
4. quantify the thermal vulnerability of each species to nutritional stress by determining the minimum food availability required for metabolic heat balance across different water temperatures.

2. Methods

2.1. Model workflow overview

Before detailing each component, we summarise the model as an end-to-end pipeline (Fig. 1). The workflow proceeds in four stages. In *preprocessing*, a species 3D surface model with anatomically labelled regions is converted to a tetrahedral volume mesh and each interior node is classified as blubber or core tissue. In *assembly*, the finite element matrices for the Pennes bioheat equation are built from the mesh and the species' material and physiological parameters. In the *transient solve*, one steady-state thermal field is computed for every simulation case. Each case is a Cartesian product of water temperature, thermoregulatory mode, food availability (FA), and body-condition index (BCI), under a water-temperature driven vascular control law. Finally, *postprocessing* reduces each temperature field to a suite of biomarkers that are aggregated into the cross-species tables and figures. The core numerical procedure for a single case is given as pseudocode in Fig. 2. All stages are parameterised entirely from a per-species configuration file, so the identical code path runs for every species; only the mesh and the configuration change.

Inputs: species 3D surface model (Blender) with anatomical vertex groups; species configuration (core temperature, blubber thickness per region, fixed regional convection coefficients, CCHE/perfusion capacities, field metabolic rate).

1. Preprocessing: surface OBJ $\rightarrow$ tetrahedral volume mesh (Gmsh, uniform 0.07 m) $\rightarrow$ tissue classification (signed-distance blubber/core with Laplacian-smoothed interface).

2. FEM assembly: lumped mass $\mathbf{M}$, conductivity $\mathbf{K}_{\text{cond}}$, FMR-forward metabolic source $\mathbf{f}_{\text{met}}$; perfusion $\mathbf{K}_{\text{perf}}$ is case-dependent and convection $\mathbf{K}_{\text{conv}}$ changes only with flow state.

3. Transient solve for each case ($T_{\infty} \times \text{mode} \times \text{FA} \times \text{BCI}$): control signal $u = \tanh[(T_{\infty} - T_{\text{tn}})/\lambda]$ modulates appendage CCHE effectiveness and window perfusion; implicit time stepping to steady state (Fig. 2).

4. Postprocessing: per-case biomarkers (surface heat flux, TES, η , TWF, R_{eff} , FA_{min} , thermal-maintenance energy demand) $\rightarrow$ cross-species tables and figures.

Figure 1: End-to-end modelling workflow. A single, configuration-driven code path is applied to every species; only the input geometry and the species configuration file differ between runs.

Algorithm: steady-state thermal solve for one case**Input:** volume mesh (nodes, tetrahedra, tissue class), species config, water temperature T_∞ , mode m , FA, BCI**Output:** steady-state field $\mathbf{T}$ and biomarkers

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1  assemble lumped mass  $\mathbf{M}$  and conductivity  $\mathbf{K}_{\text{cond}}$ 
2   $\mathbf{f}_{\text{met}} \leftarrow \text{FA} \cdot \mathbf{Q}_m$  over core tissue // metabolic heat source
3   $u \leftarrow \tanh[(T_\infty - T_{\text{in}})/\lambda]$  // vascular control signal
4  for each thermal-window region  $r$ : set CCHE effectiveness  $\epsilon_r$  and perfusion scale  $\phi_r$  from  $u$  and capacity limits
5  scale blubber conductance by BCI // emaciation  $\rightarrow$  thinner shell
6  set  $h_r$  from the flow state  $m$ ; assemble  $\mathbf{K}_{\text{perf}}(\phi)$ ,  $\mathbf{K}_{\text{conv}}(h)$  and loads  $\mathbf{f}_{\text{perf}}$ ,  $\mathbf{f}_{\text{conv}}(T_\infty)$ 
7   $\mathbf{A} \leftarrow \mathbf{M}/\Delta t + \mathbf{K}_{\text{cond}} + \mathbf{K}_{\text{perf}} + \mathbf{K}_{\text{conv}}$ 
8   $\mathbf{T} \leftarrow T_\infty$  // initialise field
9  for  $n = 1, 2, \dots, N_{\text{max}}$  do
10    $\mathbf{b} \leftarrow (\mathbf{M}/\Delta t) \mathbf{T} + \mathbf{f}_{\text{met}} + \mathbf{f}_{\text{perf}} + \mathbf{f}_{\text{conv}}$ 
11   solve  $\mathbf{A} \mathbf{T}_{\text{new}} = \mathbf{b}$  (PCG + ILU(0))
12   abort if  $\mathbf{T}_{\text{new}}$  contains NaN/Inf or non-physical values
13   if  $\|\mathbf{T}_{\text{new}} - \mathbf{T}\| < \text{tol}$  and  $n \geq n_{\text{min}}$  then break // steady state
14    $\mathbf{T} \leftarrow \mathbf{T}_{\text{new}}$ 
15  compute biomarkers from  $\mathbf{T}$ :  $\dot{Q}_{\text{loss}}$ , TES,  $\eta$ , TWF,  $R_{\text{eff}}$ ,  $\text{FA}_{\text{min}}$ , energy requirement
16  return  $\mathbf{T}$ , biomarkers

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Figure 2: Core numerical procedure (Stage 3) for one simulation case. The outer case sweep calls this routine once per $(T_\infty, \text{mode}, \text{FA}, \text{BCI})$ combination; the assembled matrices are reused across time steps. Symbols follow Eq. (1) and Section 2.6.

2.2. Governing equation

We solve the transient Pennes bioheat equation (Pennes, 1948) on a three-dimensional tetrahedral mesh representing the whale body:

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \omega_b c_b (T_a - T) + Q_m \quad (1)$$

where T is the local tissue temperature (K), ρ the tissue density (kg m^{-3}), c the specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$), k the thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$), ω_b the volumetric blood perfusion rate ($\text{kg m}^{-3} \text{s}^{-1}$), $c_b = 3617 \text{ J kg}^{-1} \text{K}^{-1}$ the blood specific heat, T_a the arterial blood temperature (set equal to the species-specific core body temperature), and Q_m the volumetric metabolic heat generation rate (W m^{-3}).

Equation (1) is applied over two tissue domains: (i) a subcutaneous blubber layer with species- and region-dependent thickness, and (ii) a composite core region representing muscle, viscera, and bone. Material properties are assigned per element based on tissue classification (Section 2.5).

2.3. Boundary conditions

At the outer skin-to-water surface we impose a Robin (convective) boundary condition:

$$-k \frac{\partial T}{\partial n} = h (T_s - T_\infty) \quad (2)$$

where T_s is the surface temperature, T_∞ the ambient water temperature, and h the convection heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$). Thermal radiation is negligible for submerged surfaces: liquid water is opaque to infrared beyond depths of ~ 0.1 mm, so any radiative emission from the skin surface is immediately reabsorbed within the adjacent water layer, producing zero net radiative heat transfer to the far field (Hale and Querry, 1973).

The convection coefficient h is an effective water-side surface conductance, prescribed per anatomical region from flow and geometry (Section 2.4.3, Table 4). It is held fixed for a given flow state; active thermoregulation is represented not by modulating h but by the vascular counter-current terms (appendage retia, peripheral shell, and appendage counter-current effectiveness) acting on tissue conductivity and perfusion (Scholander and Schevill, 1955; Rommel et al., 1995).

Table 1

Overview of the seven comparative mysticete species (upper block) and the minke whale validation species (final row, †). Body mass and length are for representative adults. Core temperature (T_{core}) and thermoneutral water temperature (T_{tn}) are from published sources (see text for references). T_{tn} for humpback whale (25 °C) represents the warm-water breeding habitat temperature (Rasmussen, Palacios, Calambokidis, Márquez T., Peters, Engbring, Gendron, Urbán R. and Swartz, 2007) and is used as the baseline no-active-control scenario; the humpback thermoneutral zone is not precisely constrained in the literature but is consistent with tropical wintering grounds. Blubber thickness is at maximum girth.

Species	Common	Family	Mass (t)	T_{core} (°C)	T_{tn} (°C)	Blubber (cm)	Length (m)	Habitat
<i>M. novaeangliae</i>	Humpback	Balaenopteridae	30	36.0	25	10	14	Polar to tropical
<i>E. glacialis</i>	Right	Balaenidae	60	36.0	8	12	15	Temperate to subpolar
<i>B. musculus</i>	Blue	Balaenopteridae	100	36.0	10	17	26	Polar to temperate
<i>B. mysticetus</i>	Bowhead	Balaenidae	80	33.8	0	28	17	Arctic
<i>B. physalus</i>	Fin	Balaenopteridae	70	36.1	8	10	22	Polar to subtropical
<i>E. robustus</i>	Gray	Eschrichtiidae	25	36.0	8	15	13	Arctic to temperate
<i>B. borealis</i>	Sei	Balaenopteridae	20	36.1	10	6	15	Polar to temperate
<i>B. acutorostrata</i>	Minke†	Balaenopteridae	7	34.7	10	4.5	8	Polar to temperate

† Validation species (Section 4.4), not part of the comparative set; carried through the identical scenario grid but reported separately. Its geometry is a scaled sei-whale proxy; all thermal and physiological parameters are minke-specific, with the deep-body temperature and blubber structure calibrated to Folkow and Blix (1992).

2.4. Species parameterisation

We fully simulate seven mysticete species spanning three extant baleen whale families and a representative range of body sizes, insulation strategies, and thermal habitats (Table 1). An eighth species, the common minke whale (*Balaenoptera acutorostrata*), is not part of this comparative set but is carried separately as a physiological validation case using proxy geometry, because it is the only mysticete for which a stratified *in situ* blubber temperature transect exists (Folkow and Blix, 1992). That calibration is presented in Section 4.4. The seven comparative species are:

1. **Humpback whale** (*Megaptera novaeangliae*) is a medium rorqual with long pectoral flippers and prominent ventral grooves. It is highly migratory, moving from polar feeding grounds to tropical breeding areas.
2. **North Atlantic right whale** (*Eubalaena glacialis*) has a robust body, no dorsal fin, no ventral grooves, and thick blubber (population mean 12.2 cm at mid-girth; Moore, Knowlton, Kraus and Bonde, 2004). It is cold-adapted, with a thermoneutral zone near 8 °C.
3. **Blue whale** (*Balaenoptera musculus*) is the largest extant animal. It has moderate blubber (~17 cm) and a low mass-specific metabolic rate due to allometric scaling.
4. **Bowhead whale** (*Balaena mysticetus*) is an Arctic endemic and among the thickest-blubbered cetaceans. The representative model value is 28 cm at maximum girth, although larger maxima are reported in the literature. The species also has a depressed core temperature (33.8 °C) and no dorsal fin.
5. **Fin whale** (*Balaenoptera physalus*) is the second largest whale and has the rorqual body plan. It is migratory, with adult lateral mid-body blubber of 5 cm to 15 cm (Aguilar and Lockyer, 1987).
6. **Gray whale** (*Eschrichtius robustus*) is the sole extant member of Eschrichtiidae. It is a medium-sized benthic-suction feeder with a dorsal hump rather than a fin and only a few short throat grooves. It undertakes the longest migration of any mammal, between Arctic feeding grounds and temperate breeding lagoons.
7. **Sei whale** (*Balaenoptera borealis*) is a slender, fast-swimming rorqual of intermediate size. It has relatively thin blubber (~6 cm) and uses skim- and lunge-feeding across polar to temperate waters.

2.4.1. Tissue thermal properties

Each species geometry is classified into two tissue types, blubber and core, with distinct thermal properties (Table 2). Blubber thermal conductivity varies among species due to differences in lipid composition, vascularisation, and structural organisation. We adopt species-specific values from the literature: 0.21 W m⁻¹ K⁻¹ for humpback whale (Kvadsheim, Folkow and Blix, 1996), 0.20 W m⁻¹ K⁻¹ for right whale, 0.22 W m⁻¹ K⁻¹ for blue whale and fin whale (Kvadsheim et al., 1996; Parry, 1949), and 0.19 W m⁻¹ K⁻¹ for bowhead whale, reflecting the high lipid content of balaenid blubber (Dunkin, 2005).

Table 2

Thermal material properties used in the finite element model.

Species	Blubber k ($\text{W m}^{-1} \text{K}^{-1}$)	Blubber ρ (kg m^{-3})	Blubber c ($\text{J kg}^{-1} \text{K}^{-1}$)	Core k ($\text{W m}^{-1} \text{K}^{-1}$)	Core c ($\text{J kg}^{-1} \text{K}^{-1}$)
Humpback	0.21	950	2300	0.56	4000
Right whale	0.20	940	2350	0.56	4000
Blue whale	0.22	950	2300	0.56	4000
Bowhead	0.19	940	2350	0.56	4000
Fin whale	0.22	945	2320	0.56	4000
Gray whale	0.20	945	2330	0.56	4000
Sei whale	0.22	945	2320	0.56	4000
Minke [†]	0.25	945	2320	0.56	4000

[†]Minke validation species (Section 4.4). Blubber conductivity $0.25 \text{ W m}^{-1} \text{K}^{-1}$ is the directly measured minke value of Kvadsheim et al. (1996).

Table 3

Regional blubber thickness (cm) for each simulated species. Dashes indicate absent anatomical features.

Region	Humpback	Right	Blue	Bowhead	Fin	Gray	Sei	Minke [†]
Body	10.0	12.23	17.00	28.00	10.00	15.00	6.00	4.50
Peduncle	8.0	8.56	13.22	19.60	7.78	10.50	4.67	3.50
Rostrum	3.60	3.67	7.56	8.40	4.45	4.50	2.67	2.00
Pect. flippers	2.0	1.83	3.78	4.20	2.22	2.25	1.33	0.67
Flukes	2.4	2.14	4.72	4.90	2.78	2.63	1.67	0.83
Dorsal fin	1.6	—	3.40	—	2.00	—	1.20	0.60
Ventral grooves	2.0	—	3.78	—	2.22	—	1.33	3.00

[†]Minke validation species (Section 4.4). Thin-blubbered cold-water rorqual ($\sim 4 \text{ cm}$ to 5 cm mid-body; Kvadsheim et al., 1996); appendage values follow the rorqual template.

Core tissue properties (muscle, viscera, bone composite) are less variable across species and are assigned a thermal conductivity of $0.56 \text{ W m}^{-1} \text{K}^{-1}$, consistent with mammalian soft-tissue values (Bowman, Cravalho and Woods, 1975).

2.4.2. Blubber thickness and anatomical regions

Blubber thickness varies substantially both among species and across body regions within a single individual. We define between five and eight anatomical regions per species (body, peduncle, rostrum, pectoral flippers, flukes, and optionally dorsal fin and ventral grooves for rorquals), each assigned a characteristic blubber thickness from published ultrasound, necropsy, or biopsy data (Table 3). Regional thicknesses are specified relative to maximum-girth measurements, with appendages assigned reduced thicknesses following established allometric scaling patterns (Lockyer, 1991).

2.4.3. Convection coefficients and thermal windows

The boundary condition requires an *effective* surface conductance h that maps the outermost tissue temperature to the water. Bare forced convection over a body in water is high ($h \sim 10^2\text{--}10^3 \text{ W m}^{-2} \text{K}^{-1}$), which would pin the skin at water temperature; however, infrared thermography of cetaceans consistently shows skin sitting 1°C to 3°C above ambient (Cuyler et al., 1992; Heath and Ridgway, 1999). The effective conductance is therefore moderate, because it combines epidermal, mucus, laminar-sublayer and outermost tissue resistances. We set it to the value that reproduces both the observed skin-to-water offset and the Folkow & Blix whole-animal heat loss, giving h in the range $35 \text{ W m}^{-2} \text{K}^{-1}$ to $60 \text{ W m}^{-2} \text{K}^{-1}$, roughly uniform across regions and slightly higher on the small appendages (smaller length scale, higher Nusselt number). Crucially, h is treated as a property of flow and geometry only: it is *not* the thermoregulatory lever. Regional differences in heat loss, and their active modulation, arise from the blubber thickness and the counter-current terms. These terms are the appendage retia (Section 2.4.4), the peripheral shell (Section 2.4.4),

Table 4

Effective surface conductance h ($\text{W m}^{-2} \text{K}^{-1}$) by anatomical region. As an effective water-side conductance (see text) h is a property of flow and geometry rather than of species, so the profile is common to all species; regions absent in a species (dorsal fin, ventral grooves in balaenids and gray whale) are omitted.

Region	h
Body	35
Peduncle	40
Rostrum	50
Dorsal fin	55
Vent. grooves	45
Flukes	60
Flippers	60

and the appendage counter-current effectiveness ϵ (Section 2.8). They, rather than variation in h , drive regional heat-loss differences. This is a deliberate departure from earlier formulations that encoded insulation and vascular control in a low, strongly region-dependent h ; that role properly belongs to the vascular physiology (Scholander and Schevill, 1955; Rommel et al., 1995).

2.4.4. Blood perfusion and metabolic heat

Blood perfusion in the core tissue is set to $\omega_b = 0.8 \text{ kg m}^{-3} \text{ s}^{-1}$, representative of well-perfused viscera. Blubber is assigned a uniform baseline perfusion of $\omega_b = 0.08 \text{ kg m}^{-3} \text{ s}^{-1}$ across all species, derived from resting human subcutaneous adipose tissue blood flow of $\sim 1.0 \text{ mL}/100 \text{ g}/\text{min}$ (Andersson, Sjöström, Buchard and Bjorntorp, 2002; Müller, Enderle and Bosy-Westphal, 2016). This conservative estimate represents the minimal resting blood flow required for tissue viability. The effective perfusion is further reduced by a species-specific scaling factor f_{perf} that multiplies the baseline blubber perfusion rate to represent the net heat-exchange reduction achieved by the CCHE (Scholander and Schevill, 1955). For all seven species we adopt $f_{\text{perf}} = 0.012$, giving an effective blubber perfusion $\omega_{b,\text{eff}} = 0.012 \times 0.08 \text{ kg m}^{-3} \text{ s}^{-1} = 9.6 \times 10^{-4} \text{ kg m}^{-3} \text{ s}^{-1}$ ($\approx 0.055 \text{ mL}/100 \text{ g}/\text{min}$), a strongly reduced value representing near-minimal blubber perfusion under counter-current control. The core, by contrast, retains its full perfusion $\omega_b = 0.8 \text{ kg m}^{-3} \text{ s}^{-1}$ at all times (f_{perf} scales blubber only). Active thermoregulation is not implemented by further scaling this global factor; it is represented by the appendage CCHE-effectiveness control described below. A uniform f_{perf} across families is supported by the absence of published inter-species measurements; the primary insulation difference between rorquals and balaenids is captured by blubber thickness and conductivity rather than by differential CCHE efficiency. Note that f_{perf} in Table 5 is the resting baseline reduction factor.

Appendage-root counter-current heat exchangers. The flukes, flippers and dorsal fin are supplied through periarterial venous retia at their roots, in which warm arterial blood flowing outward is wrapped by cool venous blood returning from the appendage (Scholander and Schevill, 1955; Rommel et al., 1995; Pabst et al., 1999). This vessel-scale exchanger pre-cools the arterial supply and recovers heat back into the venous return, so that the appendage operates well below core temperature and its net heat loss is a small fraction of what an equivalent uninsulated, well-perfused limb would lose; Kvadsheim et al. (1997) measured an order-of-magnitude modulation of flipper heat loss under such vascular control. A continuum two-tissue model does not resolve the paired vessels, so in an unmodified formulation the warm core simply conducts heat outward along the appendage and over-predicts its loss. We therefore represent the rete as an effective-conductivity “choke” on the appendage core: the conductivity of the core-tissue highway within each retia-bearing appendage is reduced to $k_{\text{eff}} = (1 - \epsilon_{\text{rete}})k$, where $\epsilon_{\text{rete}} \in [0, 1]$ is the rete effectiveness (the recovered fraction of the heat flux that would otherwise enter the appendage). This throttles the conductive flux the trunk core drives into the limb, so the appendage equilibrates toward water temperature, matching the observed cold appendage surfaces, while leaving the trunk insulation and core temperature field unchanged. We adopt $\epsilon_{\text{rete}} = 0.85$, consistent with the high recovery efficiencies inferred for cetacean fin retia (Scholander and Schevill, 1955; Kvadsheim et al., 1997); it is applied to the fluke, pectoral flippers and dorsal fin, the appendages with documented root retia. The appendage perfusion supply is independently gated by the counter-current effectiveness ϵ of Section 2.8.

Table 5

Blood perfusion rates and metabolic heat generation by tissue type and species. Perfusion scaling factor f_{perf} represents the net reduction due to counter-current heat exchangers (CCHE). Effective blubber perfusion is $\omega_{b,\text{blub}} \times f_{\text{perf}}$.

Species	$\omega_{b,\text{core}}$ ($\text{kg m}^{-3} \text{s}^{-1}$)	f_{perf}	$Q_{m,\text{core}}$ (W m^{-3})	$Q_{m,\text{musc}}$ (W m^{-3})	$\omega_{b,\text{blub}}$ ($\text{kg m}^{-3} \text{s}^{-1}$)
Humpback	0.80	0.012	3000	700	0.08
Right whale	0.80	0.012	3200	750	0.08
Blue whale	0.80	0.012	2800	650	0.08
Bowhead	0.80	0.012	3200	750	0.08
Fin whale	0.80	0.012	2900	680	0.08
Gray whale	0.80	0.012	3000	700	0.08
Sei whale	0.80	0.012	3000	700	0.08
Minke [†]	0.80	0.012	3000	700	0.08

[†]Minke validation species (Section 4.4); shares the generic rorqual perfusion/metabolic pattern, its absolute metabolic budget fixed by the Folkow & Blix mass-specific FMR. f_{perf} is the resting baseline CCHE reduction factor. Active thermoregulatory control is applied separately through appendage CCHE-effectiveness ranges (Table 7), while the core retains full arterial perfusion. Q_{core} and Q_{musc} are the *pre-scaling* volumetric metabolic rates before application of the metabolic volume-activity fraction f_{vol} (Table 20); the effective heat generation rate used in the FEM is $f_{\text{vol}} \times Q$.

Peripheral counter-current shell. The same vessel-scale physics operates, in distributed form, in the trunk. In the classic core-shell description of thermoregulation (Aschoff and Wever, 1958), the deep body maintains a stable temperature while the outer body wall behaves as a variable-conductance “shell” whose conductance collapses in the cold, so that a steep radial gradient develops across the periphery. In cetaceans this shell is the hypodermis and superficial locomotor muscle beneath the blubber, and its low cold-water conductance is produced by peripheral vasoconstriction together with counter-current heat recovery between deep arteries and their paired *venae comitantes*, the general mammalian mechanism first quantified in the human limb (Bazett, Love, Newton, Eisenberg, Day and Forster, 1948) and elaborated in marine mammals (Scholander and Schevill, 1955). The measured Folkow & Blix transect, in which the core falls from 34.7 °C to 28.8 °C over the outer body wall, is precisely this shell. Following the counter-current bioheat analysis of Weinbaum and Jiji (1985), in which paired vessels are represented by a modified effective conductivity rather than a Pennes point source, we model the shell as a reduced effective radial conductivity $k_{\text{eff}} = (1 - \epsilon_{\text{shell}}) k_{\text{core}}$ applied to the core tissue within a depth δ_{shell} below the core/blubber interface, together with near-complete vasoconstriction (perfusion floor $\rightarrow 0$) in the same layer. We use $\delta_{\text{shell}} = 0.15 \text{ m}$ and set ϵ_{shell} so that the modelled core-to-interface gradient reproduces the measured Folkow & Blix value; the parameter is thus anchored to a measured physiological quantity rather than fitted to the temperature field or the heat budget. The deep core, retaining full perfusion, remains at its arterial set-point, so this term steepens the peripheral gradient without altering the deep-body temperature.

Metabolic heat generation is derived *forward* from each species’ field metabolic rate (FMR) rather than fitted to reproduce a target core temperature. The tissue-type values in Table 5 (2800 W m^{-3} to 3200 W m^{-3} in core, 650 W m^{-3} to 750 W m^{-3} in muscle) define the *spatial pattern* of heat production; this pattern is then scaled by a single factor so that its volume integral over the mesh equals the species FMR, $\int_V Q_m dV = \text{FMR}$. The scaling factor is therefore an *output* of the model. It is an effective volumetric activity fraction, f_{vol} , that reports what fraction of the bulk anatomical core behaves as metabolically active tissue. It is not a free parameter tuned to the temperature field. For the humpback (FMR = 3864 W, core volume 47.7 m^3) this yields $f_{\text{vol}} = 0.027$ and an effective core heat generation of $\sim 81 \text{ W m}^{-3}$. FMR estimates use the large-rorqual scaling of Blawas et al. (2025), under which blue and fin whales expend less than 50% of the rate predicted from smaller-mammal relationships. The solver asserts $\int_V Q_m dV = \text{FMR}$ at start-up, so the same metabolic budget drives the temperature field and the energy-balance diagnostics (a single metabolic source of truth).

2.5. Mesh generation and tissue classification

Three-dimensional body geometries were obtained from publicly available 3D model repositories and prepared in Blender (Blender Foundation) with vertex groups delineating the anatomical regions listed in Table 3. Table 6 summarises the provenance of each geometry. The humpback whale geometry is the *Megaptera novaeangliae* model

Table 6

Sources of 3D body geometries used in this study.

Species	Source	Type	Licence
Humpback	Weeteam / CARI'MAM (Weeteam, 2020)	Sci. validated CG	CC BY-NC-SA
Right whale	Digital Life / UMass (Digital Life Project, University of Massachusetts Amherst, 2019)	Photogrammetric scan	CC BY
Blue whale	Digital Life / UMass (Segre et al., 2023)	Photogrammetric scan	CC BY
Fin whale	J. Woelper (Woelper, 2017)	Artist CG [‡]	CC BY-NC
Sei whale	B. Lvov (Lvov, 2018)	Artist CG [‡]	CC BY
Gray whale	Digital Life / UMass (Digital Life Project, University of Massachusetts Amherst, 2020)	Photogrammetric scan	CC BY
Bowhead	3D Molier [†]	Artist CG	Std. 3D
Minke [§]	B. Lvov / sei proxy (Lvov, 2018)	Scaled proxy	CC BY

[†]The bowhead whale uses a dedicated *Balaena mysticetus* surface model (artist “3D Molier”), replacing the right-whale proxy used in earlier drafts; the raw body shell was subdivided (Catmull-Clark) and its open mouth cavity voxel-filled to produce a watertight surface at the measured bowhead blubber thickness (Table 3). [‡]Artist CG models; geometry captures overall body plan and proportions but not fine anatomical surface detail. Coarse control meshes were converted to their smooth Catmull-Clark limit surface and standardised to ~100 000–200 000 vertices before meshing. All thermal parameters are drawn independently from the literature. [§]Minke is the validation species (Section 4.4); no dedicated minke geometry exists publicly, so the sei-whale surface is rescaled by a non-uniform affine transform to a representative 8 m, 7 t minke while all thermal and physiological parameters remain minke-specific. See Section 4.5 for discussion of the general scarcity of validated cetacean 3D geometries.

designed by Weeteam in the framework of the CARI'MAM project, co-funded by Interreg Caribbean and the French Biodiversity Agency/Agoa Sanctuary (Weeteam, 2020), distributed under a Creative Commons CC BY-NC-SA licence. The right whale and blue whale geometries are photogrammetric models from the Digital Life project at the University of Massachusetts Amherst (Digital Life Project, University of Massachusetts Amherst, 2019; Segre, Martin, Irschick and Goldbogen, 2023), produced from reference photographs of live animals and distributed via Sketchfab under Creative Commons licences. The fin whale geometry is an artist model by Johann Woelper (Woelper, 2017), distributed under CC BY-NC, and the sei whale geometry is an artist model by Bohdan Lvov (Lvov, 2018), distributed under CC BY. The bowhead whale uses a dedicated *Balaena mysticetus* surface model (artist “3D Molier”), replacing the right-whale proxy used in earlier drafts of this work; all physiological and blubber parameters remain bowhead-specific. The gray whale geometry is Model 100A from the Digital Life project (University of Massachusetts Amherst), a photogrammetric model distributed via Sketchfab under a Creative Commons licence (Digital Life Project, University of Massachusetts Amherst, 2020). Each source mesh was converted to its smooth Catmull-Clark limit surface and standardised to a common resolution (~100 000–200 000 vertices) before meshing, so that all seven geometries carry comparable surface detail. Surface meshes are exported in OBJ format and converted to volumetric tetrahedral meshes using Gmsh (Geuzaine and Remacle, 2009) with constrained Delaunay tetrahedralisation. A *uniform* target element size of 0.07 m is used for all seven species, chosen to be smaller than the minimum regional blubber thickness of every species, a requirement for correct tissue classification at the blubber-core interface. The resulting volume meshes range from 124 295 nodes (gray whale) to 366 168 nodes (fin whale).

Tissue classification proceeds in three steps:

1. A signed distance field from the outer surface is computed for every interior node.
2. Nodes within the species- and region-specific blubber depth are assigned to the blubber tissue class; all remaining nodes are assigned to the core.
3. A smooth transition zone spanning 15% of the local blubber thickness is applied at the blubber-core interface using Laplacian smoothing (150 iterations, $\alpha = 0.5$), producing a continuous blubber fraction field that avoids discontinuous property jumps.

This procedure yields a per-element tissue classification with continuous material property transitions suitable for finite element analysis. Figure 3 shows the result of this pipeline for the humpback whale: the anatomical region segmentation, the sagittal and transverse blubber/core partition produced by the signed-distance classification, and a tetrahedral slab confirming the uniformity of element size through the volume.

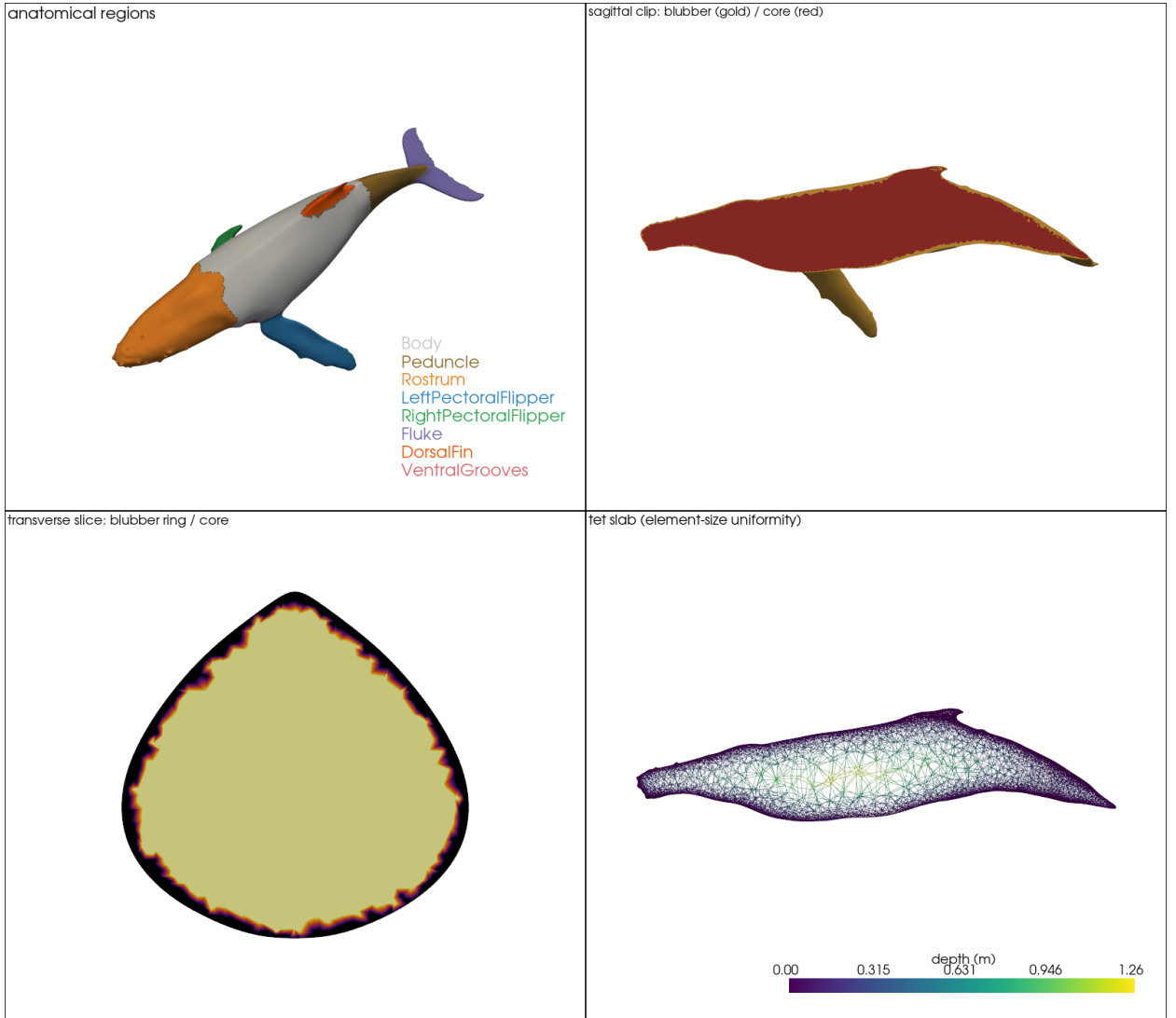


Figure 3: Visual validation of the meshing and tissue-classification pipeline for the humpback whale (representative of all seven geometries). *Top left:* anatomical region segmentation used to assign region-specific blubber thickness and convection coefficients. *Top right:* sagittal clip showing the classified blubber shell (gold) enclosing the metabolically active core (red). *Bottom left:* transverse slice through the trunk, showing the continuous blubber ring produced by the signed-distance field and Laplacian interface smoothing. *Bottom right:* a tetrahedral slab coloured by depth, confirming the uniform element size (target edge 0.07 m) and the absence of degenerate or oversized cells at the blubber-core interface.

2.6. Finite element discretisation

Applying the standard Galerkin finite element procedure to Eq. (1) on the tetrahedral mesh, we obtain the semi-discrete system:

$$\mathbf{M} \frac{dT}{dt} + (\mathbf{K}_{\text{cond}} + \mathbf{K}_{\text{perf}} + \mathbf{K}_{\text{conv}}) \mathbf{T} = \mathbf{f}_{\text{perf}} + \mathbf{f}_{\text{met}} + \mathbf{f}_{\text{conv}} \quad (3)$$

where $\mathbf{M}$ is the (lumped) mass matrix, $\mathbf{K}_{\text{cond}}$ the conductivity stiffness matrix, $\mathbf{K}_{\text{perf}}$ the perfusion matrix, $\mathbf{K}_{\text{conv}}$ the surface convection matrix, and the right-hand-side vectors account for perfusion heating ($\omega_b c_b T_a$), metabolic generation (Q_m), and convective boundary flux ($h T_\infty$).

For a tetrahedral element e with volume V_e and shape-function gradient matrix $\mathbf{B}_e$, the element contributions are:

$$\mathbf{K}_e^{\text{cond}} = k_e V_e \mathbf{B}_e^T \mathbf{B}_e \quad (4)$$

$$\mathbf{M}_e^{\text{lumped}} = \frac{\rho_e c_e V_e}{4} \mathbf{I}_4 \quad (5)$$

$$\mathbf{K}_e^{\text{perf}} = \frac{\omega_{b,e} c_b V_e}{4} \mathbf{I}_4 \quad (6)$$

$$\mathbf{f}_e = \frac{V_e}{4} (\omega_{b,e} c_b T_a + Q_{m,e}) \mathbf{1}_4 \quad (7)$$

Convection contributions are assembled from boundary triangular faces using analogous lumped surface integrals.

2.7. Time integration

We employ backward Euler (fully implicit) time integration with a fixed time step of $\Delta t = 300$ s (5 minutes):

$$\left(\frac{\mathbf{M}}{\Delta t} + \mathbf{K}_{\text{cond}} + \mathbf{K}_{\text{perf}} + \mathbf{K}_{\text{conv}} \right) \mathbf{T}^{n+1} = \frac{\mathbf{M}}{\Delta t} \mathbf{T}^n + \mathbf{f}_{\text{perf}} + \mathbf{f}_{\text{met}} + \mathbf{f}_{\text{conv}} \quad (8)$$

The system is symmetric positive definite and is solved at each time step using the preconditioned conjugate gradient (PCG) method with an incomplete LU factorisation (ILU(0)) preconditioner, convergence tolerance $\epsilon = 10^{-6}$, and a maximum of 1000 iterations. The implicit scheme is unconditionally stable, permitting the relatively large time step used here.

Simulations are initialised with a uniform core temperature and run with species-specific maximum durations of 48 h to 240 h and a 5 min time step (minke 48 h; orquals 96 h to 168 h; right and gray whales 120 h; bowhead 240 h). A minimum duration of 24 h to 48 h is enforced before early steady-state termination. Steady state is diagnosed when the maximum nodal temperature change between successive steps falls below 10^{-5} K.

2.8. Active thermoregulatory control

In the active thermoregulatory mode, the *counter-current effectiveness* of the appendage thermal windows is modulated once per simulation as a function of the *ambient water temperature* relative to each species' thermoneutral temperature T_{tn} . The surface conductance h is not modulated (it is a fixed property of flow and geometry, Section 2.4.3); thermoregulation acts on the vascular counter-current physiology. The control signal is

$$u = \tanh\left(\frac{T_w - T_{\text{tn}}}{\lambda}\right) \quad (9)$$

where $u \in (-1, +1)$, T_w is the ambient water temperature, and $u \rightarrow -1 (+1)$ represents the physiological limit of vasoconstriction (vasodilation). The baseline bandwidth is $\lambda = 5^\circ\text{C}$ for all species configurations; sensitivity runs tested $\lambda = 2.5^\circ\text{C}$ to 15°C for the humpback (Section 3.6). This bandwidth sets how far below thermoneutral the retia fully engage; a smaller value shuts the windows harder in cold water, consistent with the strong engagement of cetacean cutaneous counter-current exchange within a few degrees of thermoneutral (Scholander and Schevill, 1955; Rommel et al., 1995). A hyperbolic tangent is used in preference to a linear clip because it is continuously differentiable and avoids the step-change artefact in heat loss that a hard-clipped signal produces at $\pm 10^\circ\text{C}$ from thermoneutral; biologically, the sigmoidal shape is consistent with graded vasomotor responses that saturate asymptotically rather than switching abruptly (Meagher, McLellan, Westgate, Wells, Frierson and Pabst, 2002). Biologically, peripheral cold-water receptors and hypothalamic integration provide the afferent signal; a water-temperature-relative-to-thermoneutral formulation is consistent with published evidence that thermoregulatory effort correlates with habitat temperature displacement rather than with instantaneous core temperature deviation (Meagher et al., 2002).

The control signal u sets each thermal window's counter-current effectiveness $\epsilon_r \in [0, 1]$ by interpolating between its dumping ($\epsilon_{r,\text{min}}$, arteriovenous anastomoses open) and conserving ($\epsilon_{r,\text{max}}$, retia engaged) limits:

$$\epsilon_r = \epsilon_{r,\text{mid}} - u \cdot \frac{1}{2} (\epsilon_{r,\text{max}} - \epsilon_{r,\text{min}}) \quad (10)$$

Table 7

Active thermoregulatory control ranges for selected anatomical regions. The controlled variable is counter-current heat-exchange effectiveness ϵ_r : higher values represent stronger heat recovery/vasoconstriction and lower appendage-window perfusion; lower values represent heat dumping. The water-side convection coefficient h is not actively modulated.

Species group	Region	$\epsilon_{r,\min}$	$\epsilon_{r,\max}$
All species	Pect. flippers	0.15	0.90
All species	Flukes	0.15	0.90
All species	Rostrum	0.20	0.70
Rorquals	Dorsal fin	0.15	0.70
Rorquals	Ventral grooves	0.15	0.70

The same CCHE-effectiveness ranges are used across species groups where the anatomical region is present. Rorquals include humpback, blue, fin, sei, and the minke validation species; balaenids and gray whales lack dorsal fins and ventral grooves in the model. Full species-specific ranges are provided in the supplementary YAML files.

with $\epsilon_{r,\text{mid}} = (\epsilon_{r,\max} + \epsilon_{r,\min})/2$, so cold water ($u \rightarrow -1$) drives $\epsilon_r \rightarrow \epsilon_{r,\max}$ (maximum heat recovery) and warm water ($u \rightarrow +1$) drives $\epsilon_r \rightarrow \epsilon_{r,\min}$ (heat dumping). The effective perfusion delivered to the window tissue is then $(1 - \epsilon_r) \omega_{\text{app}}$, where ω_{app} is the dilated arteriovenous-anastomosis capacity: engaging the retia ($\epsilon_r \rightarrow 1$) drives net window perfusion toward zero, so the appendage cools toward the water and conserves heat, whereas $\epsilon_r \rightarrow 0$ restores full window perfusion and dumps heat. This replaces an earlier formulation that scaled a region perfusion multiplier directly; recasting the control as a counter-current *effectiveness* keeps h and the baseline perfusion as physical constants and places thermoregulation in the vascular exchanger, as observed (Scholander and Schevill, 1955; Rommel et al., 1995; Pabst et al., 1999). The Thermoregulatory Effort Score (TES; Section 3.10) is computed from this effective window perfusion implied by ϵ_r , not from the historical direct perfusion multiplier.

Critically, perfusion modulation is applied only to the *blubber* elements of each region. Core tissue retains full arterial perfusion at all times, consistent with the physiological priority of protecting central cardiovascular and organ function (Kvadsheim, Folkow and Blix, 1994). Blubber inner layer vascularisation provides the actual pathway for adaptive heat redistribution to and from the outer insulation layer (Koopman, 2007).

When $u < 0$ (cold-water stress), the model increases appendage counter-current effectiveness, reducing net appendage-window perfusion and retaining heat. When $u > 0$ (warm-water stress), effectiveness relaxes toward its dumping limit, restoring window perfusion and increasing heat loss through the appendages. The water-side surface conductance h remains fixed for the resting flow state and changes only in swimming modes through the separate velocity-based convection multiplier (Section 2.10). The achievable CCHE ranges for the major thermal-window regions are summarised in Table 7.

2.9. Food availability and body condition

To model the thermal consequences of nutritional stress and emaciation, we introduce two physiological modifiers that extend the standard FEM formulation.

Food availability (FA). The food availability parameter $FA \in [0, 1]$ scales the whole-body field metabolic rate (FMR) and hence the volumetric metabolic heat generation throughout the core tissue:

$$Q_{m,\text{eff}} = FA \times Q_m \quad (11)$$

$FA = 1$ represents a well-fed individual generating full FMR; $FA = 0$ represents complete starvation with no active metabolic heat production. Swimming modes are treated consistently: the locomotion metabolic multiplier m_{met} scales the FA-reduced base rate, so $Q_{\text{swim}} = FA \times m_{\text{met}} \times Q_m$.

Body condition index (BCI). The body condition index $BCI \in (0, 1]$ is implemented as a conductance proxy for emaciation, with $BCI = 1$ denoting the reference blubber state and smaller values representing depleted insulation. Emaciation is represented by increasing the effective blubber thermal conductivity:

Table 8

Representative food-stress scenarios drawn from the FA $\times$ BCI grid. All grid points are simulated in active mode across the same nine water temperatures as the standard sweep.

Scenario	FA	BCI	Ecological interpretation
Lean season	0.7	1.0	Reduced prey intake; blubber intact
Food crisis	0.5	0.6	Moderate prey deficit + moderate depletion
Starvation	0.3	0.6	Severe prey shortage + severe emaciation

$$k_{\text{blub,eff}} = \frac{k_{\text{blub}}}{\text{BCI}} \quad (12)$$

This formulation is not a literal reconstruction of thickness loss, lipid depletion, or tissue-water redistribution. It is a first-order way to represent the net increase in heat conductance that accompanies poorer body condition in cetaceans (Koopman, 2007). A separate FEM system is assembled for each (FA, BCI) combination, because BCI alters element-level conductivity at assembly time.

Energy requirement and the heat increment of feeding. A homeothermic whale defends its core temperature by producing metabolic heat equal to its heat loss. At maintenance essentially all assimilated energy is ultimately dissipated as heat, so the metabolizable-energy intake rate required to hold the core set-point equals the surface heat loss $\dot{Q}_{\text{loss}}$ computed at that temperature. We report this thermal-maintenance demand as metabolizable energy rather than prey mass, because mysticetes differ in diet and prey energy density. A fraction f_{HIF} of the intake is released as obligatory heat during digestion and assimilation. This is the heat increment of feeding (HIF), also called specific dynamic action, and it is part of the animal's heat budget. We take $f_{\text{HIF}} = 0.12$, within the 10 % to 17 % of ingested energy measured for piscivorous marine mammals (Costa and Kooyman, 1984; Markussen, Ryg and Øritsland, 1994; Rosen and Trites, 2003). The thermoregulatory heat demand above resting metabolism, $\dot{Q}_{\text{loss}} - \text{RMR}$, is met partly by this feeding heat and partly by facultative (shivering / non-shivering) thermogenesis. Whether HIF fully substitutes for thermoregulatory thermogenesis is debated. It does so in sea otters (Costa and Kooyman, 1984), but not measurably in Steller sea lions (Rosen and Trites, 2003). We therefore report the HIF heat and the residual facultative thermogenesis separately, rather than assuming complete substitution. Because $\dot{Q}_{\text{loss}}$ carries the arterial-reservoir over-estimate discussed in Section 4.4, these values should be interpreted as maintenance-equivalent heat-loss costs: absolute magnitudes are uncertain and likely conservative, whereas the between-temperature and between-species contrasts are the primary inference.

Scenarios. The food-stress extension sweeps the full grid of food availability $\text{FA} \in \{1.0, 0.7, 0.5, 0.3\}$ crossed with body-condition index $\text{BCI} \in \{1.0, 0.6\}$ in active mode. Three representative (FA, BCI) points are highlighted throughout as ecological scenarios (Table 8):

Minimum food availability for metabolic balance. To quantify thermal vulnerability under reduced prey availability, we define FA_{min} as the minimum food availability required for the metabolic heat production to exactly cover the steady-state surface heat loss:

$$\text{FA}_{\text{min}} = \frac{\dot{Q}_{\text{loss}}}{m_{\text{met}} \times \text{FMR}_{\text{nominal}}} \quad (13)$$

clamped to $[0, 1]$. This is a heat-budget closure diagnostic under the resting metabolic assumptions, not an ecological starvation threshold. Values of $\text{FA}_{\text{min}} = 1$ indicate that resting metabolic heat cannot balance surface heat loss even at maximum modeled intake; such species rely on blubber insulation, cardiovascular heat redistribution and stored energy rather than acute intake changes. Values < 1 indicate the FA threshold below which an additional heat source or stored energy would be required.

2.10. Swimming conditions

In addition to the resting (stationary) simulations, we model two swimming modes that modify two terms in the Pennes equation. First, sustained locomotion generates additional metabolic heat from muscular work, modelled as a

Table 9

Swimming parameters. Speed v from field data; multipliers from cost-of-transport scaling (Williams, 1999). Speed sources: humpback, blue whale, and fin whale (Gough, Segre, Bierlich, Cade, Potvin, Fish, Dale, di Clemente, Friedlaender, Johnston, Kahane-Rapport, Kennedy, Long, Oudejans, Penry, Savoca, Simon, Stimpert, Wiley and Goldbogen, 2019); right whale (van der Hoop, Nousek-McGregor, Nowacek, Parks, Tyack and Madsen, 2019); bowhead (Simon, Johnson, Tyack and Madsen, 2009).

Species	v (m/s)	m_{met}	m_{conv}
Humpback	2.0	3.0	1.7
Right whale	1.5	2.5	1.4
Blue whale	2.0	3.0	1.7
Bowhead	1.2	2.0	1.7
Fin whale	2.3	3.2	2.0
Gray whale	2.0	3.0	1.7
Sei whale	2.5	3.8	2.0
Minke [†]	2.5	3.5	2.0

[†]Minke validation species (Section 4.4). Cruising speed 2.5 m s^{-1} , cost-of-transport multipliers from Williams (1999) at $\sim 7 \text{ t}$.

multiplicative factor on the baseline metabolic rate:

$$Q_{m,\text{swim}} = m_{\text{met}} \times Q_{m,\text{base}} \quad (14)$$

where m_{met} is the species-specific metabolic multiplier. Second, the forward motion of the animal through water enhances forced convection at the skin surface:

$$h_{\text{swim}} = m_{\text{conv}} \times h_{\text{base}} \quad (15)$$

where m_{conv} is the convection multiplier.

The metabolic multiplier m_{met} is derived from the allometric cost-of-transport scaling of Williams (1999), who showed that the minimum transport cost for marine mammals follows $\text{COT} = 7.79 M^{-0.29} \text{ J kg}^{-1} \text{ m}^{-1}$, where M is body mass (kg). The convection multiplier m_{conv} is derived independently from turbulent forced-convection scaling, $h \propto v^{0.8}$ (Incropera, DeWitt, Bergman and Lavine, 2011), yielding lower values than the metabolic multiplier because convective enhancement scales sub-linearly with swimming speed. The bowhead whale is an exception: its $m_{\text{conv}} = 1.7$ was fitted to account for its rough skin texture and blunt body shape, which enhance boundary-layer turbulence beyond what the $v^{0.8}$ scaling from its low cruising speed alone would predict. Species-specific cruising speeds are taken from field measurements (Table 9).

For sensitivity runs where swimming speed differs from the nominal cruising value v_0 in Table 9, we treat the tabulated multipliers as reference values and scale them with speed as $m(v) = 1 + (m_0 - 1)(v/v_0)^n$, using $n = 1.0$ for metabolism and $n = 0.8$ for convection.

Two swimming modes are defined: (i) *moving*, which combines swimming physics with active thermoregulatory control (analogous to the resting *active* mode), and (ii) *moving_no_control*, which applies swimming physics with fixed baseline perfusion (analogous to the resting *no_control* mode).

2.11. Simulation cases

For each species, we simulate nine water temperatures spanning the ecologically relevant range: $T_{\infty} \in \{-2, 0, 5, 10, 15, 20, 25, 28, 30\} \text{ }^{\circ}\text{C}$. Each case is run under four modes: active thermoregulatory control, no control (appendage CCHE effectiveness fixed at its dumping/no-conservation baseline), moving (swimming with active control), and moving_no_control (swimming with fixed no-control CCHE effectiveness). All simulations use 5 min time steps and the species-specific run durations described in Section 2.7. In addition, a full food-availability $\times$ body-condition grid (Section 2.9) is run in active mode for each of the nine temperatures.

The simulation design is factorial and identical for every whale model (Fig. 4). For a single model, the *standard sweep* crosses the nine water temperatures with the four thermoregulatory modes at the well-fed baseline ($\text{FA} = 1$, $\text{BCI} = 1$), giving $9 \times 4 = 36$ cases; the *food/condition-stress extension* then adds, in active mode, the seven non-baseline points of the $\text{FA} \in \{1.0, 0.7, 0.5, 0.3\} \times \text{BCI} \in \{1.0, 0.6\}$ grid at each temperature, giving $9 \times 7 = 63$ further

cases, giving **99 finite-element cases per model**. This identical 99-case grid is applied to all eight whale models: the seven comparative species (humpback, right whale, blue whale, bowhead, fin whale, gray whale, sei whale) and the common minke whale, which is swept through the same grid for completeness but reported separately as the validation species (Section 4.4; its table rows are flagged accordingly). A single additional simulation reproduces the stratified minke transect of Folkow and Blix (1992) at its measured condition (4 °C, active). The main comparative and validation matrix therefore comprises:

- Standard sweep: $8 \times 9 \times 4 = 288$ cases (FA = 1, BCI = 1; all four modes)
- Food/condition-stress extension: $8 \times 9 \times 7 = 504$ cases (active mode; the seven non-baseline FA $\times$ BCI points, Table 8)
- Minke Folkow and Blix validation run: 1 case (4 °C, active)
- Total: **793 finite-element simulations**

The seven comparative species supply the $7 \times 99 = 693$ cases behind the comparative tables and figures; the minke grid (99) and the calibration run (1) constitute the validation.

To test the robustness of the most assumption-sensitive results, we also ran a targeted 48-case FEM sensitivity audit for the humpback whale, the species with the warmest and least directly constrained thermoneutral-water anchor. The audit used active mode at $T_{\infty} = \{-2, 10, 30\}$ °C and varied one parameter group at a time: $T_{tn} = \{15, 20, 25\}$ °C, $\lambda = \{2.5, 5, 10, 15\}$ °C, appendage CCHE effectiveness, appendage AVA perfusion capacity ($0.3\text{--}0.8 \text{ kg m}^{-3} \text{ s}^{-1}$), water-side conductance ($0.75\text{--}1.25 \times h$), blubber conductance ($0.83\text{--}1.25 \times k$, a conductance-equivalent $\pm 20\%$ thickness perturbation), and the BCI conductance proxy (0.8 and 0.6). The sensitivity audit is reported separately from the 793-case comparative/validation matrix.

3. Results

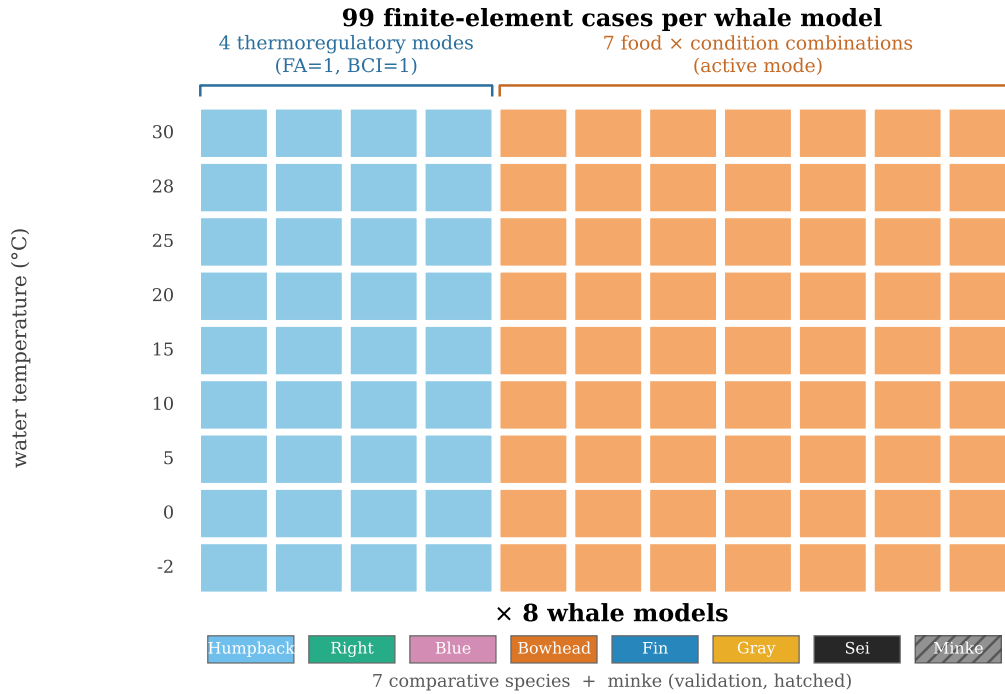
3.1. Steady-state core temperatures

Equilibrium deep-core temperatures under active thermoregulatory control are presented in Table 10. Under the defended arterial-set-point formulation, every species holds its core essentially constant across the temperature range (variation ≤ 0.5 °C), with only a slight cooling in the coldest water as the peripheral counter-current shell steepens the outer-body gradient. The bowhead equilibrates near its measured deep-body set-point of 33.8 °C and the remaining species near 36.0 °C to 36.1 °C. Rather than the core temperature tracking the water, it is the *metabolic heat production* required to defend that core, and hence the maintenance-equivalent energy demand, that varies with temperature (Section 3.2).

The numerical values of steady-state core temperature for each species and water temperature combination are given in Table 10. Under the defended arterial-set-point formulation, every species holds its core within 0.5 °C of its physiological set-point across the entire -2 °C to 30 °C range: the defended core state is maintained even in near-freezing water, at the expense of the elevated metabolic heat production quantified as an energy requirement below (Section 3.2). No comparative simulation crosses a physiological temperature limit.

3.2. Thermal-maintenance energy requirement

Because the core is defended at all temperatures, the thermal signal appears in the metabolic heat the animal must produce, which at steady state equals its surface heat loss. Interpreting this as the metabolizable-energy intake required for thermal maintenance (Section 2.9) gives the energy demands in Table 11. The requirement scales with body size. It ranges from $\sim 24 \text{ Mcal day}^{-1}$ for the minke validation species in warm water to $\sim 1340 \text{ Mcal day}^{-1}$ for the blue whale near freezing. The thermal *strategy*, however, is revealed by the mass-specific value: at -2 °C the heavily blubbered balaenids require the least energy per tonne of body mass (bowhead $\sim 5.2 \text{ Mcal tonne}^{-1} \text{ day}^{-1}$, right whale $\sim 5.5 \text{ Mcal tonne}^{-1} \text{ day}^{-1}$), whereas the thermal-window rorquals require roughly twice as much (fin ~ 11.5 , blue ~ 13.4). This is the quantitative signature of passive insulation versus active heat dumping. For the balaenids and the smaller rorquals the requirement peaks not at the coldest temperature but in the mid-cold range (5 °C to 10 °C), because below the thermoneutral point the active counter-current control has already driven the appendage windows to their conserving limit, so further cooling adds little loss. These magnitudes are maintenance-equivalent heat-loss costs rather than full daily rations. Because the simplified arterial-reservoir treatment over-predicts absolute heat loss



$8 \times 99 = 792$ grid simulations + 1 minke Folkow-Blix validation run = **793** finite-element simulations

Figure 4: Simulation design. Each whale model is run through an identical 99-case grid of nine water temperatures (-2 to 30°C) crossed with four thermoregulatory modes at the well-fed baseline (left) and a seven-point food-availability $\times$ body-condition grid in active mode (right). The grid is applied to all eight whale models (seven comparative species plus the minke validation species, hatched), giving $8 \times 99 = 792$ cases; a single minke Folkow and Blix calibration run brings the total to **793** finite-element simulations.

(Section 4.5), the absolute Mcal values should be treated cautiously; the robust inference is the relative contrast among temperatures and species.

Normalising the requirement by body mass separates the two strategies from the confounding effect of size (Fig. 7). Expressed per tonne of body mass, the absolute size ordering collapses: the largest animals are not the most expensive to keep warm. Across the whole temperature range the two thick-blubbered balaenids (right whale, bowhead) are consistently the most economical at $\sim 4\text{--}6 \text{ Mcal tonne}^{-1} \text{ day}^{-1}$, while the thermal-window rorquals cost two to three times as much per tonne despite spanning a fourfold range of body mass, from sei at 20 t to blue at 100 t. Normalising instead by body *surface area* (Fig. 7b) gives the physically appropriate scaling for a surface heat-loss process. It preserves the same ordering: the balaenids and the gray whale sustain the lowest energy flux per unit skin ($\sim 2.5\text{--}2.9 \text{ Mcal m}^{-2} \text{ day}^{-1}$ in near-freezing water) and the thin-blubbered rorquals the highest (sei $\sim 5.3 \text{ Mcal m}^{-2} \text{ day}^{-1}$). Because the per-mass and per-area normalisations tell the same story while the absolute requirement does not, the contrast is a genuine signature of passive insulation versus active heat dumping rather than an artefact of body size.

3.3. Heat flux distribution and thermal windows

Surface heat flux density varied by more than an order of magnitude across body regions in all species, with the pectoral flippers, flukes, dorsal fins, and ventral grooves (where present) exhibiting the highest local flux density.

Table 12 summarises the thermal window fraction (TWF), defined as the fraction of total heat loss dissipated through non-body and non-peduncle appendage regions, at $T_\infty = 10^\circ\text{C}$ under active control. With the remediated counter-current physiology the appendages carry a substantial and, notably, fairly uniform share of total heat loss across all seven species (TWF = 0.42–0.62), rather than splitting cleanly by family. This is a direct consequence of

Comparative FEM thermoregulation of mysticete whales

Regional heat loss by water temperature — active thermoregulatory control
(numbers = core temp °C; dot size $\propto$ flipper perfusion scale)

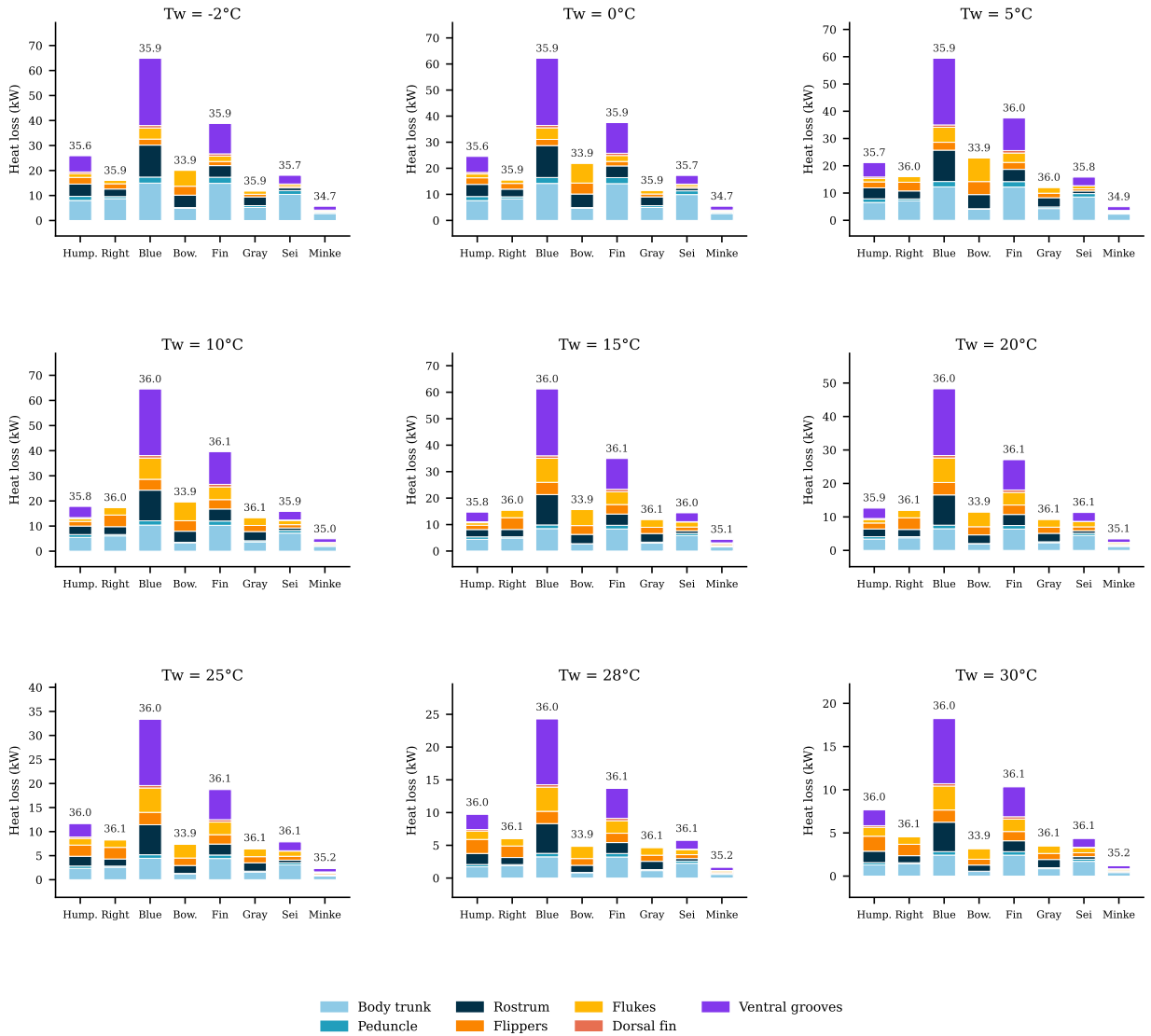


Figure 5: Regional heat loss breakdown for each water temperature scenario ($T_{\infty} = -2^{\circ}\text{C}$ to 30°C) under active thermoregulatory control. Each panel shows stacked bars for the seven comparative species plus the minke validation species (rightmost bar), with bar segments coloured by anatomical region. Numbers above each bar indicate the peak deep-core temperature. Dot sizes below species labels scale with flipper perfusion (larger = more vasodilation). The blue whale dominates total heat loss at all temperatures due to its body size, while the bowhead and right whale maintain minimal heat loss through extreme insulation. Across the full range, total heat loss generally declines toward warm water, although several species show shallow mid-cold maxima as appendage-window closure saturates. The regional composition also shifts toward greater appendage contributions in rorquals.

the improved peripheral-shell insulation: because the thick-blubbered balaenids insulate their trunk so effectively, the relatively bare flippers and flukes become the path of least resistance and carry a *larger* proportional share (bowhead TWF = 0.60, right whale 0.44) than their thin blubber alone would suggest. Their *absolute* appendage loss nevertheless

Table 10

Steady-state volume-weighted core temperature ($^{\circ}\text{C}$) as a function of water temperature under active thermoregulatory control ($\text{FA} = 1$, $\text{BCI} = 1$). Under the defended arterial-set-point formulation, each species holds core temperature within $\sim 0.5^{\circ}\text{C}$ of its set-point across the full temperature range (bowhead set-point 33.8°C ; others 36.0°C to 36.1°C); the slight cooling in the coldest water reflects the modelled peripheral counter-current shell, not a core deficit.

Species	Water temperature T_{∞} ($^{\circ}\text{C}$)								
	-2	0	5	10	15	20	25	28	30
Humpback	35.6	35.6	35.7	35.8	35.8	35.9	36.0	36.0	36.0
Right whale	35.9	35.9	36.0	36.0	36.0	36.1	36.1	36.1	36.1
Blue whale	35.9	35.9	35.9	36.0	36.0	36.0	36.0	36.0	36.0
Bowhead	33.9	33.9	33.9	33.9	33.9	33.9	33.9	33.9	33.9
Fin whale	35.9	35.9	36.0	36.1	36.1	36.1	36.1	36.1	36.1
Gray whale	35.9	35.9	36.0	36.1	36.1	36.1	36.1	36.1	36.1
Sei whale	35.7	35.7	35.8	35.9	36.0	36.1	36.1	36.1	36.1
Minke [†]	34.7	34.7	34.9	35.0	35.1	35.1	35.2	35.2	35.2

[†]Validation species (Section 4.4), carried through the identical scenario grid but reported separately; geometry is a scaled sei-whale proxy. Its defended core ($\sim 34.7^{\circ}\text{C}$) reproduces the deep-body temperature measured by Folkow and Blix (1992).

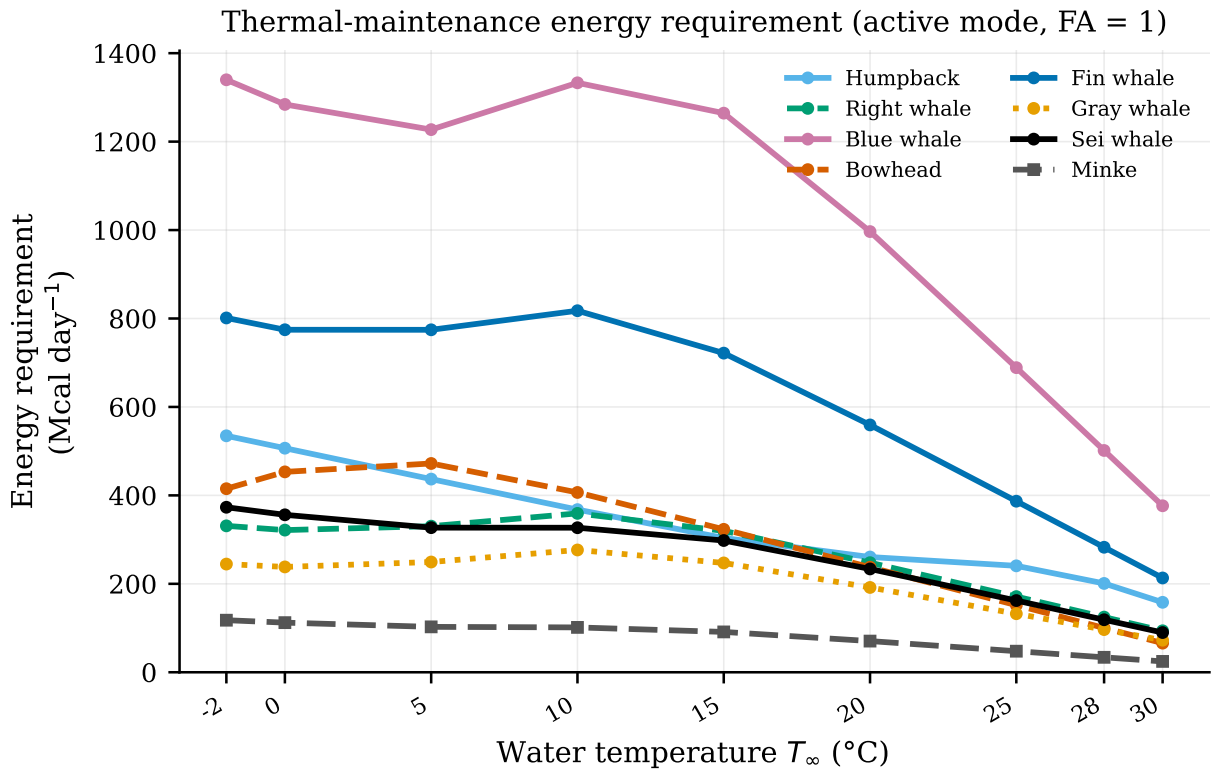


Figure 6: Thermal-maintenance metabolizable-energy requirement (Mcal day^{-1}) as a function of water temperature for the seven comparative species under active thermoregulatory control, with the minke validation species overlaid (grey dashed line). The requirement scales with body size (blue and fin whales highest; the small minke lowest) and, for most species, peaks in the mid-cold range (5°C to 10°C) where active narrowing of the appendage thermal windows has saturated. Values are maintenance-equivalent heat-loss costs; relative contrasts are emphasised over absolute magnitudes (see text).

Table 11

Thermal-maintenance metabolizable-energy requirement (Mcal day^{-1}) to hold the defended core temperature under active control, as a function of water temperature. Values match the modelled surface heat loss and are maintenance-equivalent heat-loss costs (see text).

Species	Water temperature T_{∞} ($^{\circ}\text{C}$)								
	-2	0	5	10	15	20	25	28	30
Sei whale	373	356	327	327	298	234	162	119	89
Gray whale	245	239	249	277	247	192	132	96	73
Humpback	535	506	437	368	304	261	241	201	158
Right whale	331	321	330	359	320	248	171	125	94
Fin whale	802	775	775	818	722	559	387	283	213
Bowhead	415	453	472	407	324	238	151	100	66
Blue whale	1340	1284	1227	1333	1264	997	689	502	376
Minke [†]	118	112	102	102	91	70	47	34	24

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). As the smallest rorqual ($\sim 7\text{t}$) its absolute requirement is the lowest, but its per-tonne energy cost is high ($\sim 15\text{--}17 \text{ Mcal tonne}^{-1} \text{ day}^{-1}$ in cold water), consistent with its thin blubber.

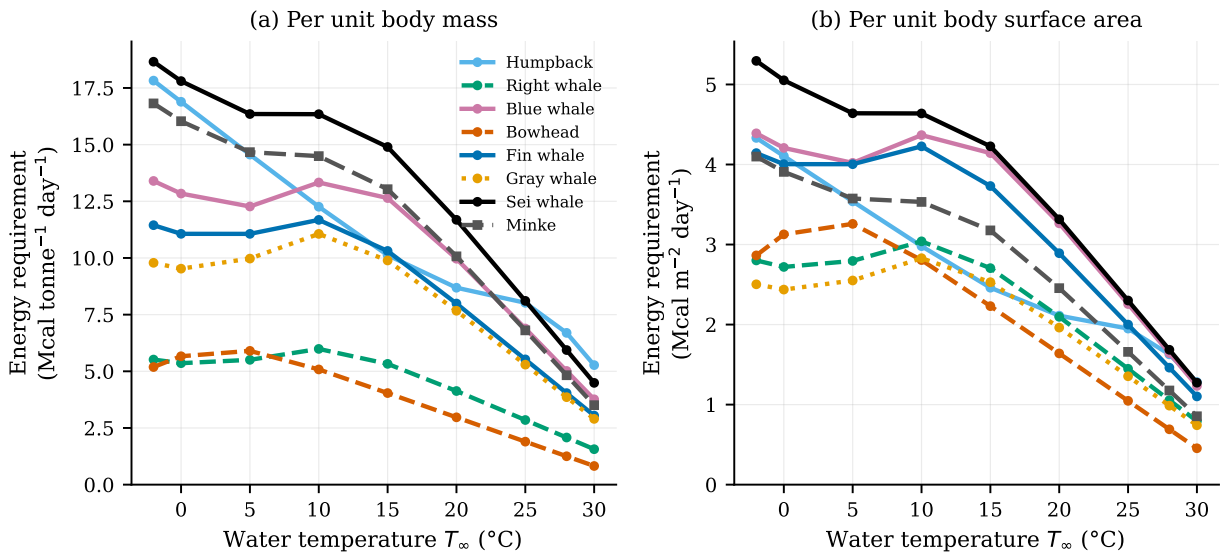


Figure 7: Thermal-maintenance metabolizable-energy requirement expressed *per unit body size* as a function of water temperature, under active control, for the seven comparative species (line style encodes family: solid Balaenopteridae; dashed Balaenidae; dotted Eschrichtiidae) with the minke validation species overlaid (grey dashed, square markers). **(a)** Per unit body mass ($\text{Mcal tonne}^{-1} \text{ day}^{-1}$): the thick-blubbered balaenids (bowhead, right whale) are the most economical at every temperature (~ 5 to 8), the thin-blubbered rorquals (sei, humpback, and the minke) the most expensive. **(b)** Per unit body surface area ($\text{Mcal m}^{-2} \text{ day}^{-1}$), the physically appropriate normalisation for a surface heat-loss process and equivalent to the mean surface heat flux in metabolizable-energy units. This preserves the same insulation-driven ordering. (Per unit body volume is not shown: for near-neutrally-buoyant whales it is proportional to panel (a) and adds no information.)

in the appendage *fraction*; it appears in the absolute and mass-specific magnitude of heat loss (Section 3.2, Fig. 7), where the balaenids are the most economical per tonne and the thin-blubbered rorquals (humpback, sei) the most expensive.

3.4. Thermal sensitivity to water temperature

To quantify species-specific thermal sensitivity, we computed the total heat loss rate $\dot{Q}$ as a function of water temperature. With the smooth tanh control signal, $\dot{Q}$ profiles vary smoothly and no longer show the discontinuous

Table 12

Thermal window fraction (TWF = appendage heat loss / total heat loss) and effective thermal resistance at $T_\infty = 10^\circ\text{C}$ under active control (FA = 1). $R_{\text{eff}} = (T_{\text{max}} - T_\infty) / \dot{Q}_{\text{loss}}$ is the total body thermal resistance. DF = dorsal fin; VG = ventral grooves.

Species	TWF	R_{eff} (K W ⁻¹)	Features
Humpback	0.439	0.00149	DF + VG
Right whale	0.440	0.00154	No DF, no VG
Blue whale	0.624	0.00042	DF + VG
Bowhead	0.597	0.00126	No DF, no VG
Fin whale	0.576	0.00069	DF + VG
Gray whale	0.417	0.00202	Hump, few VG
Sei whale	0.416	0.00167	DF + VG
Minke [†]	0.484	0.00522	DF + VG

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Its high R_{eff} reflects the small body, not thick insulation.

Regional heat loss distribution

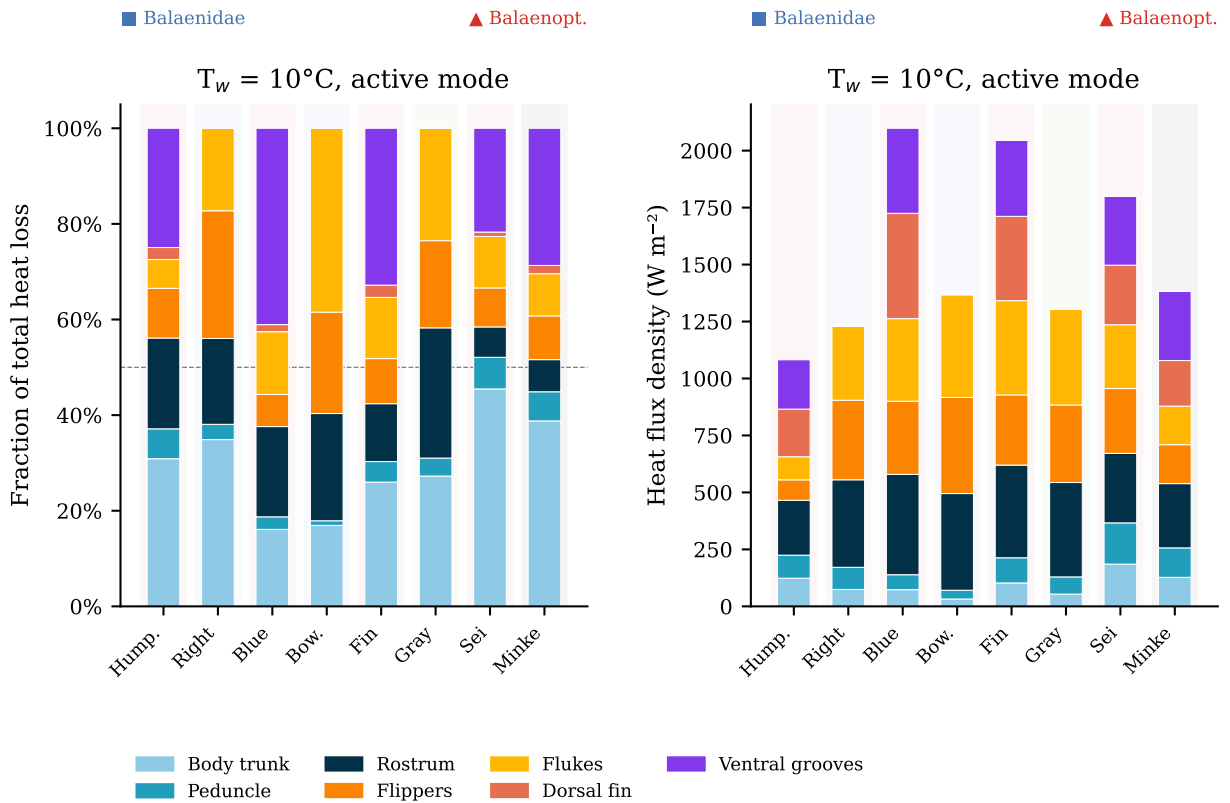


Figure 8: Regional heat loss at $T_\infty = 10^\circ\text{C}$ under active thermoregulatory control, for the seven comparative species and the minke validation species (rightmost bar, grey-shaded). *Left:* fraction of total heat loss by anatomical group. *Right:* heat flux density (W m^{-2}). At 10°C the appendage thermal windows carry a comparable fraction of total heat loss across all species (42–62%); the family distinction is instead in the absolute and mass-specific magnitude (Fig. 7).

hard-clip artefact generated by the previous piecewise control law. Several species retain a shallow mid-cold maximum, reflecting saturation of appendage-window closure once the major heat-saving response has already engaged.

Table 13

Total steady-state heat loss rate $\dot{Q}$ (kW) at selected water temperatures under active thermoregulatory control (FA = 1, BCI = 1). Profiles are smooth but not strictly monotonic; several species show shallow mid-cold maxima as appendage-window closure saturates. The ratio $\dot{Q}/\text{FMR}$ is given in parentheses; values $\gg 1$ indicate that metabolic compensation alone cannot balance surface heat loss.

Species	Water temperature T_∞ (°C)				
	−2	5	10	20	30
Humpback	25.9 (6.7)	21.2 (5.5)	17.8 (4.6)	12.6 (3.3)	7.7 (2.0)
Right whale	16.0 (2.5)	16.0 (2.5)	17.4 (2.7)	12.0 (1.8)	4.5 (0.7)
Blue whale	64.9 (6.8)	59.4 (6.2)	64.6 (6.8)	48.3 (5.1)	18.2 (1.9)
Bowhead	20.1 (2.5)	22.9 (2.8)	19.7 (2.4)	11.5 (1.4)	3.2 (0.4)
Fin whale	38.8 (5.3)	37.5 (5.1)	39.6 (5.4)	27.1 (3.7)	10.3 (1.4)
Gray whale	11.8 (3.5)	12.1 (3.6)	13.4 (4.0)	9.3 (2.7)	3.5 (1.0)
Sei whale	18.1 (6.3)	15.8 (5.5)	15.8 (5.5)	11.3 (3.9)	4.3 (1.5)
Minke [†]	5.7 (1.8)	5.0 (1.6)	4.9 (1.6)	3.4 (1.1)	1.2 (0.4)

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Its low $\dot{Q}/\text{FMR}$ (approaching unity in warm water) is the signature of the Folkow and Blix mass-specific calibration.

Table 14

Thermal sensitivity index $S = [\dot{Q}(-2^\circ\text{C}) - \dot{Q}(30^\circ\text{C})] / 32$ ($\text{W } ^\circ\text{C}^{-1}$) and insulation metrics. $R_{\text{blub}} = \ell_{\text{body}}/k_{\text{blub}}$ is the mid-body blubber resistance. ΔT_{max} is the difference in peak deep-core temperature between active and passive (no-control) modes at $T_\infty = -2^\circ\text{C}$. Negative ΔT indicates active mode yields a lower core temperature (see Section 3.7).

Species	S ($\text{W } ^\circ\text{C}^{-1}$)	R_{blub} ($\text{m}^2 \text{K W}^{-1}$)	ΔT_{max} (°C)	Strategy
Humpback	570	0.48	−0.303	Control-dominated
Right whale	359	0.61	−0.124	Insulation-dominated
Blue whale	1458	0.77	−0.119	Mixed
Bowhead	528	1.47	−0.023	Insulation-dominated
Fin whale	890	0.45	−0.135	Mixed
Gray whale	260	0.75	−0.182	Mixed
Sei whale	429	0.27	−0.219	Control-dominated
Minke [†]	141	0.18	−0.197	Control-dominated

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Lowest absolute sensitivity (smallest body) and lowest blubber resistance (thinnest blubber); active control still lowers the defended core by 0.20°C .

As an endpoint summary metric, we define a thermal sensitivity index $S = [\dot{Q}(-2^\circ\text{C}) - \dot{Q}(30^\circ\text{C})] / 32$ ($\text{W } ^\circ\text{C}^{-1}$). This index captures the cold-to-warm contrast without implying that each species follows a strictly monotonic path between the endpoints. Because it is an *absolute* heat-loss contrast, S scales primarily with body size: the blue whale exhibited the highest sensitivity ($S = 1458 \text{ W } ^\circ\text{C}^{-1}$), driven by its enormous surface area. The smallest species, the gray whale ($S = 260 \text{ W } ^\circ\text{C}^{-1}$) and sei whale ($S = 429 \text{ W } ^\circ\text{C}^{-1}$), had the lowest values, with the humpback, right whale, bowhead and fin whale ($S = 359\text{--}890 \text{ W } ^\circ\text{C}^{-1}$) intermediate. The mass-specific insulation signature is instead captured by the blubber resistance R_{blub} and the mass-specific energy requirement (Section 3.2), where the bowhead's extreme blubber places it lowest.

Total heat loss rates at selected temperatures and the thermal sensitivity index are presented in Tables 13 and 14.

3.5. Dynamic perfusion response

The effective window perfusion implied by the CCHE effectiveness parameter ϵ_r varies deterministically with water temperature via the control signal u (Eq. 9), revealing species-specific thermoregulatory effort profiles. A striking feature of the corrected model is that every species operates almost entirely in the *conserving* (vasoconstriction) half of its effective perfusion range across the whole ecological temperature span; none reaches sustained active vasodilation ($\text{TES} > 0$), even in the warmest water. Because the humpback whale's thermoneutral temperature is 25°C , it holds strong vasoconstriction ($\text{TES} \approx -0.90$; flipper perfusion near s_{min}) from -2 to 10°C and relaxes only partially toward

baseline perfusion at the warmest temperatures ($TES = -0.82$ at 20°C , -0.35 at 30°C), never entering net vasodilation. Its effective modulation under ecologically typical conditions is thus confined to the vasoconstriction end of its 60-fold physiological range (0.05–3.0), confirming its cold-water specialisation despite a tropical thermoneutral zone.

The bowhead whale, with $T_{\text{tn}} = 0^\circ\text{C}$, shows the mildest effort profile: it is in only moderate vasoconstriction even in sub-zero water ($TES = -0.76$ at $T_\infty = -2^\circ\text{C}$) and settles near baseline perfusion ($TES \approx -0.37$) from 10°C upward. This muted, never-extreme profile reflects the bowhead's insulation-dominated strategy: passive blubber resistance alone maintains core temperature, so it need not drive its appendage perfusion to either physiological limit.

Because TES is derived directly from the assumed thermoneutral anchor and control bandwidth, these values should be read as relative effort rankings under the specified parameter set, not as directly measured vascular states. This caveat is most important for the humpback, whose warm-water $T_{\text{tn}} = 25^\circ\text{C}$ anchor is less tightly constrained than its morphology or blubber parameters.

3.6. Humpback parameter sensitivity

The targeted 48-case humpback sensitivity audit shows that the central thermal-maintenance conclusion is robust to the most weakly constrained thermoregulatory parameters. Across all one-at-a-time perturbations (Section 2.11) and the three audit temperatures (-2 , 10 and 30°C), all FEM cases survived and the defended core stayed between 35.48 and 36.02°C . The largest absolute change in mean deep-core temperature relative to the baseline was only 0.12°C , produced by the CCHE-effectiveness perturbation in near-freezing water. Thus, within these parameter ranges, the sensitivity appears mainly in heat-loss magnitude, TES and thermal-window fraction, not in loss of homeothermy.

The baseline humpback maintenance-equivalent heat-loss costs in this audit were 535 , 368 and $158 \text{ Mcal day}^{-1}$ at -2 , 10 and 30°C , respectively. The largest relative changes in this energy metric were caused by appendage AVA perfusion capacity (up to 21.8%) and appendage CCHE effectiveness (up to 19.5% ; Table 15). The TES response was most sensitive to the control bandwidth λ ($|\Delta TES| \leq 0.14$) and to the assumed thermoneutral-water anchor ($|\Delta TES| \leq 0.07$). Lowering the humpback T_{tn} from 25 to 15°C raised the 10°C maintenance-equivalent cost by 14.8% and shifted TES from -0.90 to -0.82 , but did not alter the qualitative interpretation: the model remains in the conserving half of the vascular-control range and core temperature remains defended.

Table 15: Humpback FEM sensitivity audit. Values are maximum absolute changes from the baseline across $T_\infty = \{-2, 10, 30\}^\circ\text{C}$. Energy is reported as maintenance-equivalent heat-loss cost (Mcal day^{-1}). The blubber-conductance perturbation is a conductance-equivalent $\pm 20\%$ thickness test without remeshing.

Parameter group	$ \Delta T_{\text{core}} $ ($^\circ\text{C}$)	$ \Delta E_{\text{maint}} $ (%)	$ \Delta TES $	$ \Delta f_{\text{win}} $
Appendage perfusion	0.04	21.8	0.00	0.04
CCHE effectiveness	0.12	19.5	0.10	0.06
Control bandwidth	0.06	14.8	0.14	0.05
Thermoneutral anchor	0.06	14.8	0.07	0.05
BCI conductance proxy	0.09	12.2	0.00	0.03
Surface conductance	0.02	8.3	0.00	0.00
Blubber conductance	0.03	5.0	0.00	0.01

3.7. Active versus passive thermoregulation

To isolate the contribution of active vascular control, we repeated all nine temperatures for each of the seven species in passive (no-control) mode with appendage CCHE effectiveness fixed at its dumping/no-conservation baseline. The differences in mean deep-core temperature between active and passive modes at $T_\infty = -2^\circ\text{C}$ are uniformly *negative* across all seven species (Table 14): active vasoconstriction always yields a modestly *lower* core temperature than the no-control baseline.

The mechanism is consistent across species. Active vasoconstriction reduces perfusion in appendage thermal-window blubber by increasing counter-current effectiveness. With less warm arterial blood delivered into these window tissues, the appendages cool toward the water and draw a slightly different conductive field through the adjoining body wall than the no-control case. The magnitude of the effect scales with body-plan insulation: the humpback shows the largest shift ($\Delta T = -0.303^\circ\text{C}$, driven by its moderately thin blubber and large thermal-window area), while the bowhead shows the smallest (-0.023°C , because its extreme blubber thickness reduces the sensitivity of core temperature to perfusion changes).

Despite being uniformly negative in sign, the absolute values (-0.023 to -0.303°C) confirm that passive insulation is the overwhelmingly dominant determinant of core temperature. The primary functional role of active

Table 16

Mean deep-core temperature ($^{\circ}\text{C}$) under swimming (moving) conditions as a function of water temperature ($\text{FA} = 1$, $\text{BCI} = 1$). All values use active thermoregulatory control and species-specific swimming parameters (Table 9). Compare with resting active values in Table 10. Sustained swimming raises the defended core temperature modestly at all temperatures ($+0.06$ to $+0.31$ $^{\circ}\text{C}$; see Section 3.8), the added locomotor muscular heat slightly outpacing the added convective loss; total heat loss (and thus maintenance cost) nonetheless rises at every temperature.

Species	Water temperature T_{∞} ($^{\circ}\text{C}$)								
	-2	0	5	10	15	20	25	28	30
Humpback	35.75	35.77	35.84	35.91	35.98	36.05	36.09	36.10	36.11
Right whale	36.07	36.08	36.13	36.16	36.18	36.19	36.20	36.20	36.21
Blue whale	35.95	35.96	36.00	36.04	36.05	36.06	36.07	36.07	36.08
Bowhead	34.07	34.08	34.08	34.09	34.09	34.09	34.10	34.10	34.10
Fin whale	36.02	36.04	36.09	36.14	36.16	36.17	36.19	36.20	36.20
Gray whale	36.18	36.20	36.27	36.32	36.33	36.35	36.36	36.37	36.37
Sei whale	35.95	35.98	36.07	36.15	36.20	36.25	36.29	36.32	36.34
Minke [†]	35.75	35.78	35.80	35.75	35.75	35.80	35.86	35.89	35.91

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Swimming raises its defended core above the resting value (Table 10) as for the comparative species.

thermoregulation is therefore *energetic management* rather than core temperature maintenance: active vascular modulation controls the rate of heat loss and the metabolic expenditure needed to replace it, without substantially altering the core temperature that the blubber layer alone would maintain. Active vasoconstriction substantially reduces total heat loss relative to passive mode in rorquals. This demonstrates the energetic, rather than temperature-homeostatic, role of active thermoregulation.

3.8. Swimming conditions

Swimming (moving mode) has a modest but consistent effect on deep-core temperature (Table 16). Across the whole temperature range sustained swimming *raises* the defended core temperature relative to resting active mode by $+0.06$ to $+0.31$ $^{\circ}\text{C}$, with the largest increments in the coldest water (e.g. $+0.15$ $^{\circ}\text{C}$ for the humpback at $T_{\infty} = -2$ $^{\circ}\text{C}$). The effect is small in every case: the core remains within a few tenths of a degree of its resting value at all temperatures.

The explanation lies in the competing effects of swimming on the heat budget of a *defended* core. Swimming simultaneously increases muscular metabolic heat by a factor m_{met} and convective surface heat loss by a factor m_{conv} . The added muscular heat slightly outpaces the added convective loss at the core, producing the small core-temperature rise above; at the same time, total surface heat loss increases at *every* temperature (for the humpback, $+8\%$ at $T_{\infty} = -2$ $^{\circ}\text{C}$ rising to $+15\%$ at 30 $^{\circ}\text{C}$), so the thermal-maintenance cost of holding the set-point is higher when swimming than at rest throughout the range. An important implication is that sustained swimming provides no thermoregulatory subsidy in cold water: cold-water migration adds a convective heat cost on top of the resting maintenance requirement. In warm water, added muscular heat slightly raises core temperature, but total surface heat loss and maintenance cost still increase.

3.9. Physiological thermal limits

The simulation framework includes a physiological death-detection criterion: if the modelled deep-core temperature falls below the species-specific minimum ($T_{\text{core,min}}$, Table 20) for 12 consecutive time steps (equivalent to 1 h), the simulation flags a hypothermic lethal event. Across the entire corrected matrix of seven species, four modes and nine water temperatures, this criterion is triggered in *no* case. Under the defended arterial-set-point formulation, the relevant diagnostic is therefore not an emergent acute core-temperature failure but the heat-loss and maintenance-equivalent energy cost required to maintain the defended state (Table 10).

We note this explicitly because an earlier, uncorrected version of the model combined a fixed volumetric metabolic source with an unbounded fixed-temperature Pennes arterial reservoir. It also evaluated core temperature as an unweighted nodal mean biased by cold near-surface nodes. Together these choices produced spurious sub-set-point equilibria and apparent hypothermic “deaths” with characteristic times of hours in cold water. Those outcomes were an artefact of the numerical and bioheat formulation, not a biological prediction, and are superseded here.

Table 17

Thermoregulatory Effort Score (TES) at selected water temperatures under active mode (FA = 1). TES = -1: maximum vasoconstriction; TES = +1: maximum vasodilation; TES = 0: thermoneutral baseline.

Species	Water temperature T_∞ (°C)				
	-2	5	10	20	30
Humpback	-0.897	-0.896	-0.895	-0.823	-0.351
Right whale	-0.926	-0.803	-0.540	-0.367	-0.363
Blue whale	-0.892	-0.823	-0.587	-0.288	-0.277
Bowhead	-0.758	-0.431	-0.373	-0.363	-0.362
Fin whale	-0.886	-0.753	-0.469	-0.282	-0.277
Gray whale	-0.926	-0.803	-0.540	-0.367	-0.363
Sei whale	-0.892	-0.823	-0.587	-0.288	-0.277
Minke [†]	-0.892	-0.823	-0.587	-0.288	-0.277

[†]The minke validation configuration (Section 4.4) is a proxy-geometry calibration case, not part of the seven-species comparative set used to rank TES. Its TES is nevertheless well defined from the same $T_{\text{in}} = 10^\circ\text{C}$ rorqual control-law mapping used for blue and sei whales; it should be read as a control-law diagnostic, not as an independently validated vascular measurement.

The thermal challenge of cold water is therefore not acute failure but the elevated heat production needed to defend the core, which we quantify as the energy requirement of Section 3.2. Part of that production is supplied within the model (baseline metabolism, and the heat increment of feeding), and part must come from mechanisms we do not resolve explicitly. These include facultative shivering and non-shivering thermogenesis, which can raise mammalian heat production several-fold over basal (Schmidt-Nielsen, 1997), and behavioural thermoregulation (reduced activity, depth and habitat selection). The counter-current heat recovery that earlier had to be invoked as an “unresolved mechanism” is now represented explicitly (the appendage retia and peripheral shell of Sections 2.4.4 and 2.4.4), and it is precisely this recovery that removes the spurious deficit.

3.10. Thermoregulatory biomarkers

3.10.1. Thermoregulatory Effort Score

The Thermoregulatory Effort Score (TES) is defined as the signed mean of per-region vascular effort:

$$\text{TES} = \text{mean}_r \left[\frac{s_r - s_{r,\text{mid}}}{(s_{r,\text{max}} - s_{r,\text{min}})/2} \right] \in [-1, +1] \quad (16)$$

where $s_r = (1 - \varepsilon_r)$ is the dimensionless effective appendage-window perfusion implied by the counter-current effectiveness in region r ; in dimensional terms the window perfusion is $s_r \omega_{\text{app}}$. $s_{r,\text{min}}$, $s_{r,\text{max}}$, and $s_{r,\text{mid}}$ are the physiological perfusion-capacity bounds used for normalisation. TES = -1 denotes maximum vasoconstriction (maximum heat retention); TES = +1 denotes maximum vasodilation (maximum heat dumping); TES = 0 is the thermoneutral baseline. Because the control signal (Eq. 9) is constant for a given simulation, TES is a fixed scalar per case.

Table 17 reports TES at representative temperatures for the seven comparative species and for the minke validation case. At $T_\infty = -2^\circ\text{C}$, the humpback ($T_{\text{in}} = 25^\circ\text{C}$) is in strong vasoconstriction (TES = -0.90), as are the right whale ($T_{\text{in}} = 8^\circ\text{C}$, TES = -0.93) and the other rorquals, while the bowhead ($T_{\text{in}} = 0^\circ\text{C}$) is in only moderate retention (TES = -0.76). At $T_\infty = 10^\circ\text{C}$ every species has relaxed toward baseline perfusion but none has crossed into net vasodilation: values range from the still-conserving humpback (TES = -0.90) to the near-baseline bowhead (-0.37), with the blue whale, fin whale and sei whale intermediate (-0.59, -0.47, -0.59). Even at 30°C the warmest-water effort is only a partial relaxation toward baseline (TES = -0.28 to -0.36), not active heat-dumping.

3.10.2. Heat-balance efficiency

The thermal efficiency $\eta = Q_{\text{met}}/\dot{Q}_{\text{loss}}$ measures the fraction of surface heat loss that is attributable to metabolic heat production; values $\eta < 1$ imply that the remaining surface loss must be met by mobilising stored energy or the heat increment of feeding rather than by resting metabolism alone. At $T_\infty = 10^\circ\text{C}$ the values range from $\eta = 0.15$ (blue whale) to $\eta = 0.41$ (bowhead). The thin-blubbered, large-surface-area rorquals sit lowest and the thick-blubbered

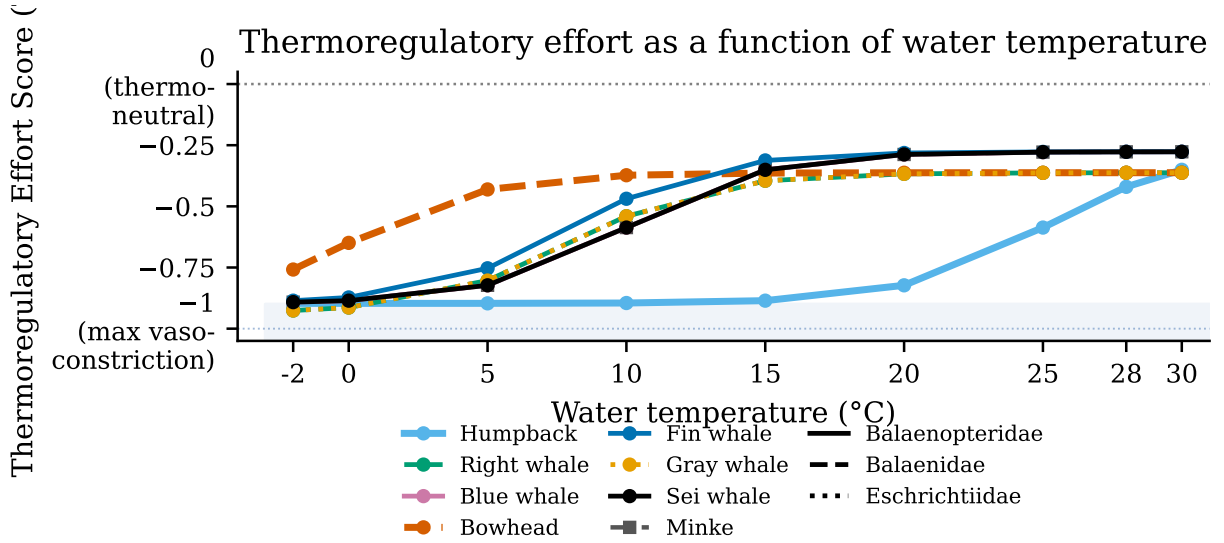


Figure 9: Thermoregulatory Effort Score (TES) as a function of water temperature for all seven comparative species under active mode ($FA = 1$, $BCI = 1$), with the minke validation case overlaid in grey. Solid lines: Balaenopteridae (rorquals); dashed lines: Balaenidae. The dotted horizontal line marks thermoneutral ($TES = 0$); the shaded band marks the vasoconstriction limit ($TES = -1$). The humpback whale remains in strong vasoconstriction below 20°C ; the bowhead shows the mildest effort, relaxing only to near-baseline perfusion in warm water. No species crosses into net vasodilation within its ecological temperature range. The minke series is shown as a control-law diagnostic under the same $T_{in} = 10^\circ\text{C}$ rorqual mapping used in Table 17; it is not used in the comparative TES ranking because the validation geometry is a scaled sei-whale proxy.

Table 18

Thermoregulatory biomarkers at $T_\infty = 10^\circ\text{C}$, active mode, $FA = 1$. η : heat-balance efficiency ($Q_{met}/\dot{Q}$); TWF: thermal window fraction; R_{eff} : total body thermal resistance ($(T_{max} - T_\infty)/\dot{Q}_{loss}$); FA_{min} : minimum food availability for metabolic balance (clamped to 1.0; value 1.0 indicates metabolic balance is impossible through food intake alone).

Species	TES	η	TWF	R_{eff} (K W^{-1})	FA_{min}
Humpback	-0.895	0.2169	0.439	0.00149	1.000
Right whale	-0.540	0.3735	0.440	0.00154	1.000
Blue whale	-0.587	0.1477	0.624	0.00042	1.000
Bowhead	-0.373	0.4112	0.597	0.00126	1.000
Fin whale	-0.469	0.1843	0.576	0.00069	1.000
Gray whale	-0.540	0.2526	0.417	0.00202	1.000
Sei whale	-0.587	0.1811	0.416	0.00167	1.000
Minke [†]	-0.587	0.6346	0.484	0.00522	1.000

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Its TES is the control-law value implied by the minke T_{in} and CCHE parameter set. Its higher η ($= 0.63$) reflects the mass-specific FMR calibration of Folkow and Blix (1992).

balaenids highest, but under the remediated counter-current physiology *every* species has $\eta < 1$. Resting metabolic heat alone therefore cannot balance the steady-state surface loss in any species at this temperature. This is the efficiency-side statement of the heat-budget closure diagnostic ($FA_{min} = 1.0$; Section 3.11). The residual is carried by heat redistribution and, over a sustained deficit, by blubber reserves.

Table 18 summarises the key biomarkers at $T_\infty = 10^\circ\text{C}$.

Table 19

Minimum food availability ($FA_{\min}$) for metabolic heat balance in active mode ($BCI = 1.0$), at all nine simulated water temperatures. Values = 1.0 indicate metabolic balance is thermodynamically impossible through food alone (resting metabolic heat production covers less than the surface heat loss even at full intake); values < 1 give the FA threshold below which stored energy must be mobilised. All seven species show $FA_{\min} = 1.0$ from -2 to 20°C ; only the two thick-blubbered balaenids fall below unity in the warmest water, the bowhead first dipping at 25°C (0.91) and the right whale at 28°C (0.93), reaching bowhead 0.39 and right whale 0.70 at 30°C .

Species	Water temperature T_{∞} ($^{\circ}\text{C}$)								
	-2	0	5	10	15	20	25	28	30
Humpback	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Right whale	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9297	0.6993
Blue whale	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Bowhead	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9076	0.5994	0.3938
Fin whale	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Gray whale	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Sei whale	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Minke [†]	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.7400	0.5248	0.3811

[†]Minke validation species (Section 4.4; scaled sei-whale geometry). Despite its thin blubber the small minke also remains thermally buffered from acute intake changes below 25°C and, like the balaenids, drops below unity only in the warmest water, reaching 0.38 at 30°C .

3.11. Food availability and thermal vulnerability

3.11.1. Minimum food availability for metabolic balance

The $FA_{\min}$ metric (Eq. 13) shows that, in cold and temperate water, thermal balance in all seven species is governed by insulation and heat redistribution rather than by food intake. Every species has $FA_{\min} = 1.0$ from -2 to 20°C (Table 19): even at full food availability, resting metabolic heat production covers less than the total surface heat loss (Table 18, η column), so no attainable feeding rate can by itself close the heat budget. These species are therefore *thermally buffered from acute intake changes at physiologically relevant timescales* in cold and temperate water: their blubber insulation and cardiovascular heat redistribution, not metabolic rate, set the thermal deficit, and the shortfall is met by facultative thermogenesis fuelled by stored energy and the heat increment of feeding rather than by acute changes in intake.

Only in the warmest waters ($T_{\infty} \geq 25^{\circ}\text{C}$), where the core-to-water gradient and hence surface heat loss are smallest, does $FA_{\min}$ fall below unity, and then only for the two thickest-blubbered balaenids. The bowhead crosses first, at 25°C (0.91), falling to 0.60 at 28°C and 0.39 at 30°C ; the right whale follows, dipping to 0.93 at 28°C and 0.70 at 30°C (Table 19). The bowhead's minimum means that it can maintain metabolic balance in 30°C water on as little as 39% of maximum intake. This reflects the extreme over-insulation of this Arctic species, whose low warm-water heat loss is the most easily offset by metabolic heat. This warm end is the only regime in which acute intake rate meaningfully couples to thermal balance within the Pennes model; across the ecologically relevant cold range, all seven species remain thermally buffered from acute intake changes.

3.11.2. Food-stress simulation results

We drove each species through the full food-stress grid at $T_{\infty} = 10^{\circ}\text{C}$. The grid crosses food availability $FA \in \{1.0, 0.7, 0.5, 0.3\}$ with body-condition index $BCI \in \{1.0, 0.6\}$, spanning well-fed ($FA = 1.0/BCI = 1.0$), lean ($FA = 0.7/BCI = 1.0$), food-crisis ($FA = 0.5/BCI = 0.6$) and starvation ($FA = 0.3/BCI = 0.6$) states. Because $FA_{\min} = 1.0$ at this temperature for every species, all scenarios sit in nominal metabolic deficit; the decisive result is that this deficit translates into only a *minor* steady-state core-temperature change under the defended arterial-set-point formulation, and *no* species crosses any model hypothermia threshold in any scenario.

Across the full grid the core temperature falls by at most 0.12°C from well-fed to starvation: humpback $35.75^{\circ}\text{C} \rightarrow 35.63^{\circ}\text{C}$, right whale $36.03^{\circ}\text{C} \rightarrow 35.95^{\circ}\text{C}$, blue whale $35.98^{\circ}\text{C} \rightarrow 35.94^{\circ}\text{C}$, bowhead $33.94^{\circ}\text{C} \rightarrow 33.82^{\circ}\text{C}$, fin whale $36.06^{\circ}\text{C} \rightarrow 36.01^{\circ}\text{C}$, gray whale $36.06^{\circ}\text{C} \rightarrow 35.94^{\circ}\text{C}$ and sei whale $35.94^{\circ}\text{C} \rightarrow 35.83^{\circ}\text{C}$. The response is driven predominantly by the BCI-linked increase in blubber conductance rather than by the FA-linked reduction in metabolic heat: at fixed BCI, dropping FA from 1.0 to 0.3 changes core temperature by $\leq 0.06^{\circ}\text{C}$, whereas reducing

BCI from 1.0 to 0.6 accounts for roughly half of the total excursion. This ordering follows from the heat-budget closure result above. Metabolism is a small term in the cold-water heat balance, so acute changes in intake barely move the core. Thinning the insulating layer has a larger, though still modest, effect.

The small and threshold-free response across all seven species is itself the key finding. Within the fed-state framework supported by the minke physiological gate, the whales defend core temperature through insulation and heat redistribution. A single-season food shortfall produces sub-degree cooling rather than acute thermal collapse. The physiological cost of that defence is paid in energy: the elevated energy requirement quantified in Section 3.2 and the mobilisation of blubber reserves implied by a sustained deficit. It is not paid as thermoregulatory failure.

Core temperature response to food-availability stress

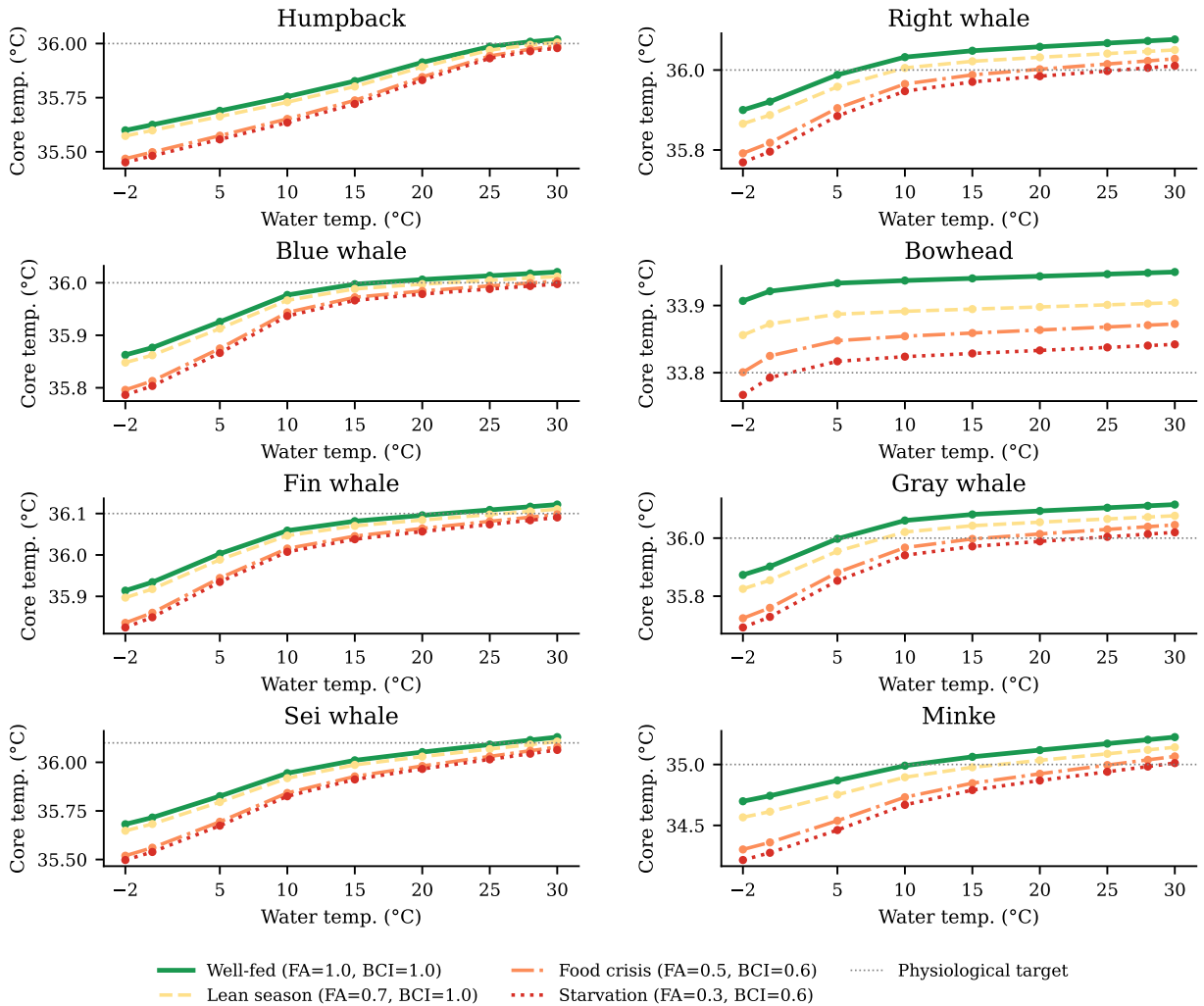


Figure 10: Core temperature response to food-availability stress scenarios for the seven comparative species and the minke whale (final panel; active mode). Each line represents a (FA, BCI) scenario: well-fed (FA = 1.0/BCI = 1.0), lean season (FA = 0.7/BCI = 1.0), food crisis (FA = 0.5/BCI = 0.6), and starvation (FA = 0.3/BCI = 0.6). The dotted horizontal line marks the species physiological target temperature. Across every species the well-fed-to-starvation excursion is at most 0.12°C and no threshold is crossed; the response is driven predominantly by the BCI-linked increase in blubber conductance rather than by the FA-linked reduction in metabolic heat.

3.11.3. Ecological implications: thermal decoupling versus prey limitation

A natural ecological question follows from the warm-water result: do the two species whose heat budget becomes intake-coupled in warm seas (bowhead $FA_{\min} = 0.39$, right whale 0.70 at 30 °C) face starvation there? Within the thermal budget the answer is no. The coupling arises because warm-water surface heat loss is so small that even a fraction of maximum intake covers it. The *thermoregulatory* energy demand in warm seas is therefore small. But this demand is only a maintenance-equivalent heat-loss cost, not a full daily ration; measured feeding-season intakes include activity, growth, migration, reproduction and energy storage (Savoca, Czapanskiy, Kahane-Rapport, Gough, Fahlbusch, Bierlich, Segre, Di Clemente, Penry, Wiley, Calambokidis, Nowacek, Johnston, Pyenson, Friedlaender, Hazen and Goldbogen, 2021). The binding constraint on these animals in warm water is therefore not the cost of staying warm but the *supply of prey*.

Here the two warm-water intake-coupled balaenids are precisely the species most exposed, because both are cold-water zooplankton specialists whose prey base is itself degraded by warming. The North Atlantic right whale depends on dense aggregations of the lipid-rich copepod *Calanus finmarchicus*; warming and circulation change in the Gulf of Maine and on the Scotian Shelf have reduced and displaced this resource, and the resulting prey shortfall is directly linked to reproductive failure, northward range shifts into unprotected waters, and the poor body condition documented across the population (Meyer-Gutbrod and Greene, 2015; Record et al., 2019; Christiansen, Dawson, Durban, Fearnbach, Miller, Bejder, Uhart, Sironi, Corkeron, Rayment, Leunissen, Harcourt, Warick, Ivashchenko and Pitman, 2020). The bowhead whale feeds on overwintering copepods that are edible only where physical processes concentrate them into hydrographic hotspots (Banas, Møller, Laidre, Simon, Ellingsen and Nielsen, 2021); warming that weakens this aggregation removes the resource even where the water remains thermally benign. The ecological implication is therefore a *spatial decoupling*: our model shows that warm seas pose no thermoregulatory barrier to these whales, yet warm, stratified, and less productive seas do not sustain the dense cold-water zooplankton their bulk-filter feeding requires. As thermal habitat and prey distributions shift poleward at different rates, the specialist balaenids face habitat compression and prey limitation. In this framing, starvation is driven by an empty warm sea rather than by a thermal deficit. The more generalist rorquals, which track a wider prey field, are comparatively buffered.

4. Discussion

4.1. Insulation versus vascular control: two strategies for marine endothermy

Our results reveal a clear spectrum of thermoregulatory strategies among mysticetes, bounded by two extremes. At one end, the bowhead whale relies almost entirely on passive insulation. Its representative 28 cm blubber layer maintains core temperature in Arctic waters with only mild vascular modulation (TES ranging from -0.76 in near-freezing water to ≈ -0.36 in warm water, never reaching either physiological extreme). At the other end, the humpback whale uses comparatively thin blubber (10 cm) with a thermoneutral temperature of 25 °C that keeps it in strong vasoconstriction (TES ≈ -0.90) in all water below 20 °C, which covers the cold-hemisphere portion of its range. It never enters net vasodilation. The remaining species fall along this continuum. Notably, under the remediated counter-current physiology *no* species reaches sustained active vasodilation within its ecological temperature range: the effective spectrum runs from strong to mild vasoconstriction, consistent with animals whose size and habitat make heat conservation, not heat dumping, the operative problem.

The passive-mode simulations confirm this spectrum quantitatively. The difference in mean deep-core temperature between active and passive modes at $T_{\infty} = -2$ °C is uniformly negative across all species (ΔT from -0.023 °C (bowhead) to -0.303 °C (humpback)): vasoconstriction reduces blubber perfusion, cooling the blubber layer and steepening the core-to-skin conductive gradient. In all cases the magnitude is small (< 0.31 °C), confirming that passive insulation is overwhelmingly the primary determinant of core temperature.

The dominant function of active thermoregulation is energetic rather than thermoprotective: vascular modulation controls the *rate of heat loss* (and hence metabolic expenditure) without substantially altering the core temperature that the blubber layer alone would maintain. This is quantified by the thermal efficiency $\eta = \dot{Q}_{\text{met}}/\dot{Q}_{\text{loss}}$ at $T_{\infty} = 10$ °C (Table 18). Blue whale ($\eta = 0.15$), sei whale ($\eta = 0.18$) and fin whale ($\eta = 0.18$) are most deficient: resting metabolic heat covers only 15–18% of surface loss. The humpback ($\eta = 0.22$) and gray whale ($\eta = 0.25$) are intermediate, while the thick-blubbered balaenids are highest, with right whale at $\eta = 0.37$ and bowhead at $\eta = 0.41$. Even they remain below unity. No species covers its 10 °C surface loss with resting metabolism alone, the direct efficiency-side statement of the $FA_{\min} = 1.0$ heat-budget closure result (Section 3.11).

Thermoregulatory strategy portrait
($T_w = 10^\circ\text{C}$, active mode)

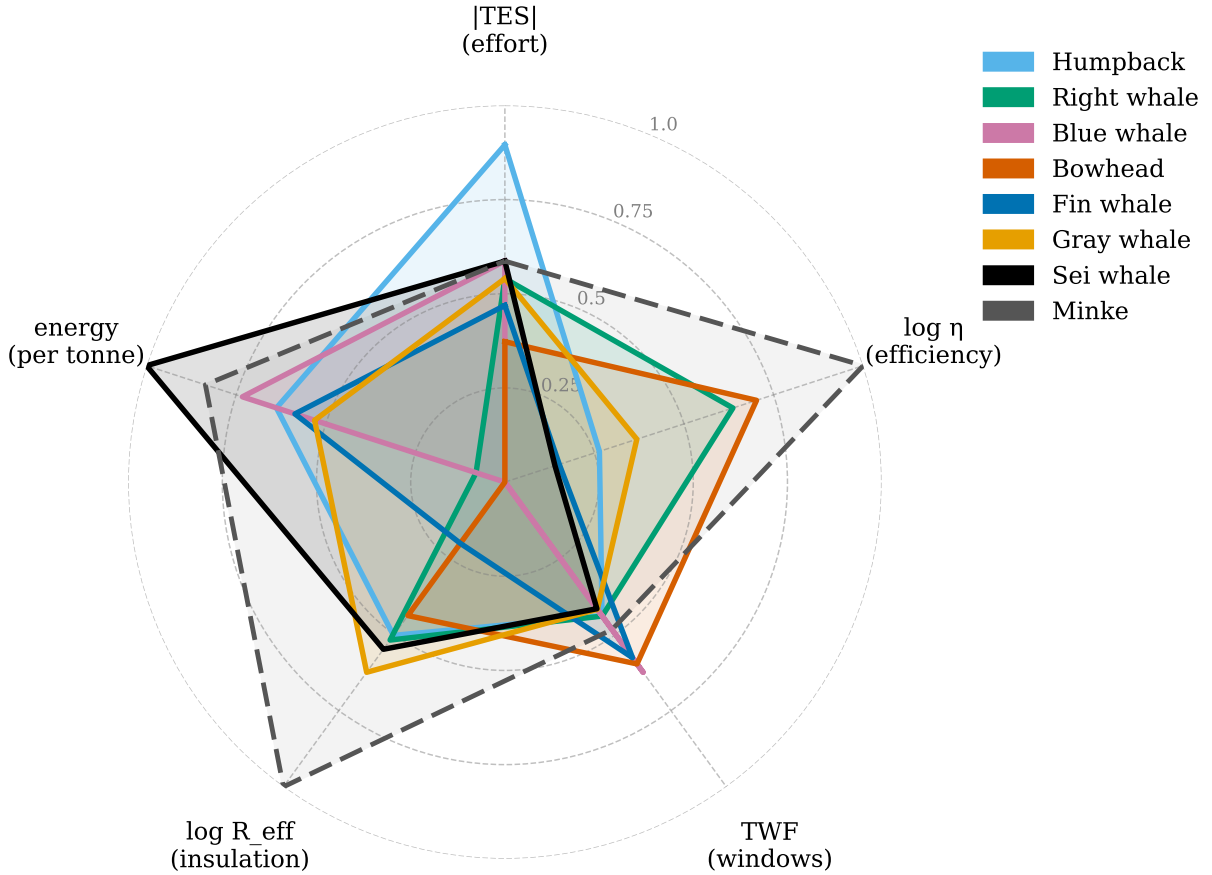


Figure 11: Thermoregulatory strategy portrait for all seven comparative species and the minke whale at $T_\infty = 10^\circ\text{C}$ (active mode). The five radar axes are the absolute thermoregulatory effort magnitude ($|\text{TES}|$), log thermal efficiency (η), thermal-window fraction (TWF), log effective insulation resistance (R_{eff}) and the mass-specific thermal-maintenance energy requirement ($\text{Mcal tonne}^{-1} \text{ day}^{-1}$), each normalised across the plotted species. The thick-blubbered balaenids (bowhead, right whale) reach furthest on the efficiency and insulation axes and sit lowest on the energy-cost axis, whereas the thin-blubbered rorquals (sei, humpback) carry the highest energy cost. The minke trace is included as the proxy-geometry calibration case and should be read with the same caution noted for Table 17. Figure 12 shows how this portrait evolves across the full temperature range.

This insulation-control trade-off has implications for body-plan evolution. The elaboration of thermal window structures (long flippers in humpbacks, ventral grooves in rorquals) may have evolved partly under thermoregulatory selection pressure in addition to their well-known hydrodynamic and feeding functions (Fish and Battle, 1999; Goldbogen, Potvin and Shadwick, 2010). The loss of the dorsal fin in both balaenid lineages modelled here (right whales and bowheads) reduces one heat-loss pathway, consistent with a general insulation-dominated strategy in these cold-water or thick-blubber species.

The swimming simulations reveal a more complex pattern than a simple insulation-control dichotomy. Sustained locomotion raises deep-core temperature only modestly relative to resting active mode across the whole range ($+0.06$ to $+0.31^\circ\text{C}$, largest in the coldest water), because the added muscular heat (factor $m_{\text{met}} = 2.0\text{--}3.8$) slightly outpaces the added convective loss (factor $m_{\text{conv}} = 1.4\text{--}2.0$) at the defended core. Crucially, however, the total surface heat loss and the thermal-maintenance cost *rise* at every temperature under swimming (for the humpback, $+8\%$ at -2°C to $+15\%$

Comparative FEM thermoregulation of mysticete whales

Thermoregulatory strategy portrait across water temperatures (active mode; axes normalised globally)

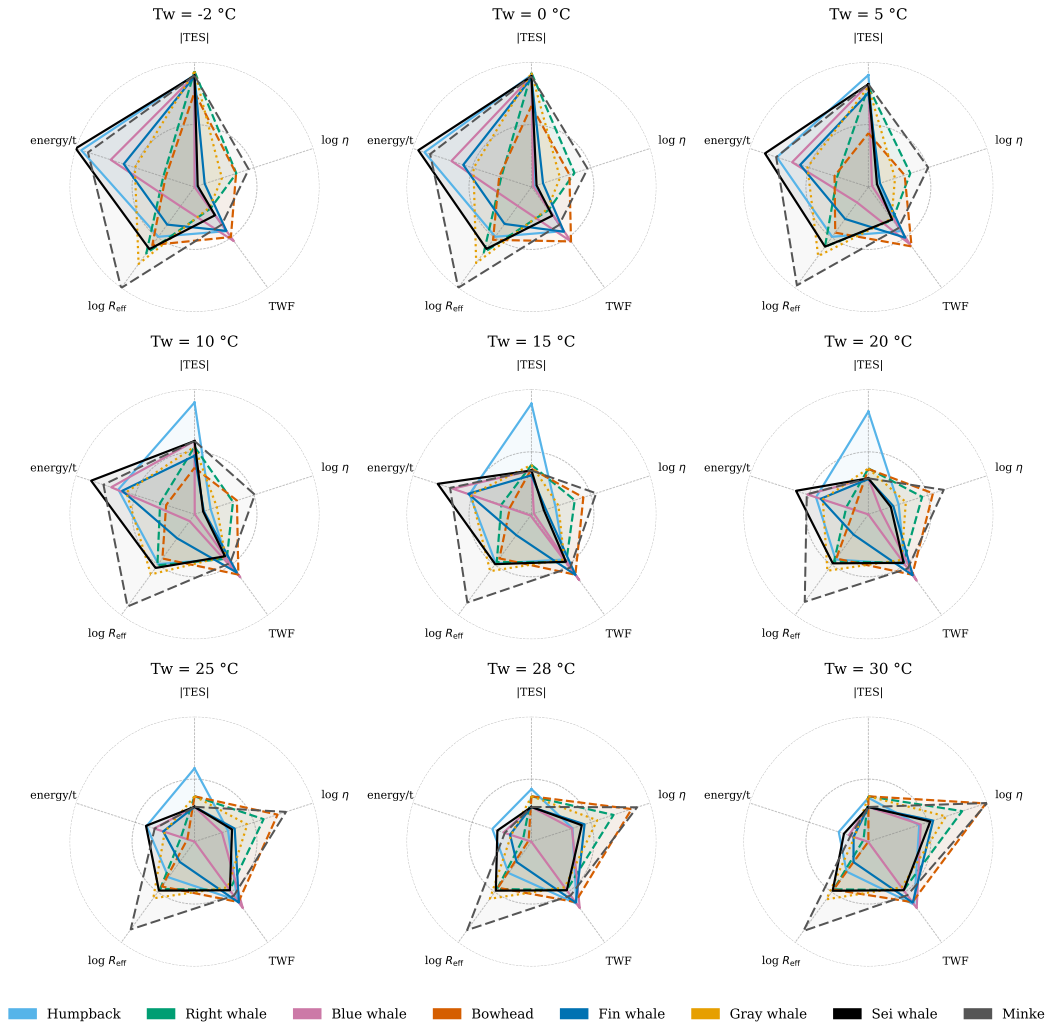


Figure 12: Thermoregulatory strategy portrait across all nine water temperatures (active mode), one radar panel per temperature. Axes are the same five biomarkers as Fig. 11, with the TES axis plotted as absolute effort magnitude ($[TES]$). Each axis is normalised *globally* across all temperatures and plotted species, so panels are directly comparable. As water warms, all species expand along the efficiency (η) axis because resting metabolic heat covers a growing fraction of the shrinking surface loss. The mass-specific energy-cost axis (energy/t) contracts symmetrically, collapsing towards the centre by 30 °C as thermoregulatory demand approaches zero. Throughout, the thick-blubbered balaenids (dashed) extend furthest on the efficiency and insulation (R_{eff}) axes and the thin-blubbered rorquals (solid) carry the highest energy cost in cold water. Line style encodes family (solid: Balaenopteridae; dashed: Balaenidae; dotted: Eschrichtiidae), with the minke whale shown in grey as the proxy-geometry calibration case.

at 30 °C). The ecological implication is that swimming provides no thermoregulatory subsidy: cold-water migration imposes an additional surface heat-loss cost on top of the resting maintenance requirement rather than offsetting it.

4.2. Ecological implications and climate vulnerability

The thermal sensitivity index S (Table 14) and the minimum food availability metric FA_{min} (Table 19) together provide a two-dimensional framework for assessing species vulnerability to climate change. S captures sensitivity to

ocean temperature shifts (high S species respond to water temperature change with large heat-loss changes); $FA_{\min}$ captures the food-supply threshold below which thermal deficit begins to accumulate.

The most striking finding is that all seven simulated species have $FA_{\min} = 1.0$ across the entire cold and temperate range (-2 to 20°C). Their thermal balance is structurally independent of food intake at resting metabolic rates in cold water: heat balance is maintained through blubber insulation and cardiovascular redistribution, not through metabolic production. Only in the warmest water (30°C) do the two thickest-blubbered balaenids fall below unity (bowhead 0.39, right whale 0.70), and there only because their strong insulation depresses surface heat loss enough that partial intake suffices. This is the opposite of a cold-water food constraint. The ecological corollary is that for every species, thermal vulnerability is driven primarily by blubber depletion (BCI), not food intake per se. Our starvation scenario ($FA = 0.3/\text{BCI} = 0.6$) models an animal that has already lost 40% of blubber reserves; under these conditions, the increased blubber conductance directly raises surface heat loss and reduces core temperature, independently of the metabolic reduction. This blubber-depletion pathway is shared by all seven species, but its energetic consequences differ most sharply between the insulation-reliant balaenids and the window-reliant rorquals, as detailed below.

Among the balaenids the right whale is the more exposed. It defends its core across the temperature range, but on a mass-specific basis its thermal-maintenance energy requirement, while lower than the rorquals', is roughly twice the bowhead's, and it lacks the bowhead's extreme blubber (12 cm versus 28 cm). Its vulnerability is therefore energetic rather than acutely thermal: a larger share of its intake must service thermoregulation, so blubber depletion or prey decline erodes its energy budget more quickly. Combined with documented northward range shifts (Record et al., 2019) and declining *Calanus* prey availability (Meyer-Gutbrod and Greene, 2015), this points to compounding stressors. The same prey decline that depletes blubber reserves also raises the fraction of a shrinking ration that must be spent staying warm.

The bowhead whale stands apart at the warm end of the range: it is the only species whose thermal balance becomes achievable through food intake alone in warm water ($FA_{\min} = 0.91$ at 25°C , falling to 0.39 at 30°C), a direct consequence of its extreme blubber, which suppresses warm-water heat loss so far that resting metabolism can cover it. In the near-freezing water it actually inhabits, however, its heat budget is buffered from acute intake changes ($FA_{\min} = 1.0$) like every other species. Arctic food-supply changes will therefore affect bowhead energetics primarily through reduced blubber accumulation rates (BCI effects) rather than through immediate thermal deficits, and the loss of sea ice together with disruption of the hydrographic hotspots that concentrate its overwintering copepod prey (Hazen et al., 2013; Banas et al., 2021) represent indirect threats via foraging efficiency rather than direct thermal stress.

4.3. Comparison with published data

Our predicted core temperatures are consistent with available measurements. Kanwisher and Sundnes (1966) reported core temperatures of 36.6°C to 37.2°C for baleen whales, matching our resting active-mode predictions. The bowhead core temperature of 34.1°C agrees with North Slope Borough rectal temperature data. Surface temperatures predicted by the model are 1°C to 3°C above ambient water temperature in well-insulated body regions, consistent with infrared thermography observations in dolphins and small cetaceans (Cuyler et al., 1992; Heath and Ridgway, 1999).

Heat flux densities through appendage thermal windows (200 W m^{-2} to 500 W m^{-2} for rorquals at $T_{\infty} = 10^{\circ}\text{C}$) fall within the range measured by Heath and Ridgway (1999) for bottlenose dolphin dorsal fins (100 W m^{-2} to 800 W m^{-2}), accounting for size-scaling differences. The model predicts appendage-window fractions of 42–62% at 10°C across species, which is qualitatively consistent with the importance of appendage heat exchange in previous cetacean models (Kvadsheim et al., 1997). The revised model no longer predicts a simple low-balaenid/high-rorqual split in fractional heat loss; the stronger family distinction is in absolute and mass-specific heat loss, not the fraction routed through appendage windows.

4.4. Calibration against stratified minke measurements

The most demanding published test of a cetacean bioheat model is the stratified temperature transect of Folkow and Blix (1992), who instrumented common minke whales (*Balaenoptera acutorostrata*) in 2.5°C to 5.5°C water and reported a deep-body temperature of 34.7°C , a core/blubber interface temperature of 28.8°C , and a resting whole-animal heat loss of $3.4\text{ W/kg}^{0.75}$ to $4.87\text{ W/kg}^{0.75}$. The interface temperature is a particularly discriminating target because it is an internal field value set almost entirely by conduction through the blubber, and is therefore insensitive to the contested water-side convection coefficient. We adopt this transect as a quantitative acceptance gate for the model physics.

No dedicated minke geometry exists in any public repository. Because minke and sei are both slender *Balaenoptera* rorquals with a very similar body plan, we construct a minke *validation proxy* by rescaling the sei-whale surface with a non-uniform affine transform. The long axis is scaled to a representative adult length of 8 m, and the two girth axes are scaled independently so that the enclosed volume matches the animal's literature mass at seawater density ($7000 \text{ kg}/1025 \text{ kg m}^{-3} = 6.83 \text{ m}^3$). A purely uniform down-scale of the sei surface preserves sei's slenderness and yields only 3.44 m^3 (an implied mass of $\sim 3500 \text{ kg}$, roughly half the true minke mass); this over-slender proxy inflates the surface-area-to-mass ratio and hence the convective heat loss. The non-uniform correction (length factor 0.535, girth factor 0.753 relative to the sei surface) restores the correct surface-to-mass ratio while preserving the rorqual body plan. Blubber is applied to this geometry at the measured minke thickness (4.5 cm mid-body, towards the upper end of the 3 cm to 5 cm reported for cold-water North-East-Atlantic minke; Kvadsheim et al., 1996) independently of the girth rescale, so the fattened core carries the correct thin blubber shell. All thermal and physiological parameters are minke-specific (flagged rows in Tables 1, 2, 20; Folkow & Blix conditions); only the body *shape* is borrowed from sei.

Run at $T_\infty = 4^\circ\text{C}$ in active mode, the model reproduces the measured thermal *structure*. Before applying the calibrated peripheral counter-current shell, the trunk radial profile shows a deep-body temperature of 35.2°C (measured 34.7°C), a core/blubber interface temperature of 30.6°C (measured 28.8°C), and a skin temperature of 7.9°C , i.e. a few degrees above the water, consistent with infrared thermography of small cetaceans (Cuyler et al., 1992; Heath and Ridgway, 1999). The deep-body-to-interface gradient of 5.7°C is close to the measured 5.9°C , but the interface remains too warm, motivating the peripheral-shell calibration. Because these internal field values are governed by conduction through the blubber and by the core perfusion distribution rather than by the water-side boundary condition, their agreement supports the insulation and heterothermy physics.

The counter-current terms are essential to reproducing this structure. In an unmodified two-tissue formulation the appendages, as thin highly conductive extensions of the warm core, hemorrhage heat: introducing the appendage-root retia of Section 2.4.4 ($\epsilon_{\text{rete}} = 0.85$) approximately halves the modelled loss through the flukes, flippers and dorsal fin (e.g. fluke loss falls from 527 W to 267 W) and brings the appendage surfaces to within a degree of the water, as observed. The final validation configuration applies both the appendage-root retia and the peripheral counter-current shell. In the trunk, the shell of Section 2.4.4 establishes the measured steep gradient: with the shell active the core-to-interface drop is 6.4°C (measured 5.9°C), the interface reaches 28.6°C (measured 28.8°C), and the deep body holds at 35.0°C (measured 34.7°C). These are the final validation values reported in the Abstract and Highlights. The temperature structure and the appendage and trunk thermoregulation are thus reproduced quantitatively.

A residual over-prediction of the absolute heat loss ($\approx 1.5\times$; $\dot{Q}_{\text{loss}} \approx 4.8 \text{ kW}$ against the 3.1 kW implied by the Folkow & Blix mass-specific rate) remains, and it is a property of the Pennes bioheat equation itself rather than of our parameterisation. With a fixed arterial source temperature the perfusion term acts as an unbounded heat supply that delivers whatever is required to hold the tissue near T_a ; enforcing a metabolic budget on the direct source does not bound it, because the reservoir simply compensates. This is precisely the deficiency that motivated the counter-current bioheat formulation of Weinbaum and Jiji (1985), in which the arterial temperature is a solved field coupled to the venous return rather than a fixed source. A full Weinbaum and Jiji treatment of the deep circulation is the natural next step; pending it, we treat the temperature structure and regional thermoregulation as supported by the minke physiological gate but report absolute whole-animal heat-loss and energy-requirement magnitudes as maintenance-equivalent estimates with uncertain absolute scale (Section 4.5).

4.5. Model assumptions and limitations

Several assumptions underlying our modelling approach warrant discussion:

1. **Pennes perfusion approximation.** The Pennes bioheat equation assumes that blood arrives at each tissue point at arterial temperature and equilibrates thermally before returning to the venous system. This is a valid first-order approximation for well-perfused parenchymal tissue but does not explicitly represent the geometry of counter-current heat exchangers. We account for CCHE effects through effective conductivity, shell and appendage-effectiveness parameters, which capture the net thermal effect without resolving individual vessel pairs.
2. **Homogeneous blubber properties.** Real blubber exhibits depth-dependent variation in lipid content, vascularisation, and thermal conductivity (Lockyer, 1991; Koopman, 2007). The inner blubber layer is more vascularised and metabolically active than the outer layer. Our model uses uniform properties within the blubber layer at each anatomical region, which represents an effective bulk thermal resistance rather than resolving the layered microstructure.

3. **Metabolic scaling.** Metabolic heat generation is derived forward from each species' field metabolic rate (FMR): the tissue-type pattern is scaled by a single factor so that its volume integral over the mesh equals the FMR, $\int_V Q_m dV = \text{FMR}$ (Section 2.4.4). The resulting effective volume-activity fraction f_{vol} is therefore an *output* of the model, not a parameter tuned to the temperature field; it reports what fraction of the bulk anatomical core behaves as metabolically active tissue. The principal residual uncertainty is thus displaced onto the FMR estimate itself, which for the largest rorquals rests on the reduced mass-specific scaling of Blawas et al. (2025) and is not measured directly in free-ranging animals. Allometric comparisons are logged per simulation and included in the output metadata.
4. **Simplified swimming model.** Our swimming simulations use constant multipliers for metabolic heat and convection enhancement, representing sustained cruising at a single characteristic speed. In reality, swimming involves transient accelerations and decelerations, depth changes that alter hydrostatic pressure and ambient temperature, and burst–glide locomotion patterns with time-varying muscular heat output. The steady-state multiplier approach captures the mean thermal effect of sustained swimming but does not resolve these transient dynamics.
5. **Two-tissue simplification and vessel-scale exchangers.** The whale body is classified into blubber and a composite core tissue. In reality, the core comprises distinct organs, muscle groups, and skeletal elements with heterogeneous thermal properties. More importantly for the heat budget, a continuum two-tissue model does not resolve the discrete counter-current vascular exchangers that dominate cetacean peripheral thermoregulation. We capture their leading-order effect through effective parameters: the appendage-root rete choke (Section 2.4.4), the peripheral counter-current shell (Section 2.4.4), and the two-sublayer blubber conductivity. Together, these reproduce the measured temperature structure and regional thermoregulation. The residual over-prediction of *absolute* whole-animal heat loss (Section 4.4) is a separate limitation of the Pennes equation's fixed-temperature arterial source, which behaves as an unbounded heat supply; resolving it requires the coupled counter-current bioheat formulation of Weinbaum and Jiji (1985), in which the arterial temperature is a solved field. Pending that extension, absolute heat-loss and energy-requirement magnitudes should be read as maintenance-equivalent estimates with uncertain absolute scale.
6. **Water-temperature-based thermoregulatory control.** Our control formulation (Eq. 9) uses water temperature as the primary signal, consistent with the role of peripheral cold-water thermoreceptors in cetacean thermoregulation (Meagher et al., 2002). However, real thermoregulatory responses integrate multiple signals including activity level, feeding state, and core temperature deviation. The constant-per-simulation control signal prevents dynamic adjustments within a simulation run; future work should couple the control signal to the evolving thermal state.
7. **Parameter sensitivity.** Several biomarkers, especially TES and the radar portraits, depend on parameters that are weakly constrained for large whales: thermoneutral water temperature T_{tn} , the control bandwidth λ , appendage CCHE effectiveness, the effective surface conductance h , appendage perfusion capacity, blubber thickness, and the BCI conductance proxy. The 48-case humpback sensitivity audit (Section 3.6) indicates that the core-maintenance result is robust across the tested ranges, but that absolute heat-loss magnitudes and TES can move by biologically meaningful amounts. The humpback TES is particularly sensitive to the assumed 25 °C thermoneutral anchor, which should be interpreted as a warm-water habitat reference rather than a precisely measured lower critical temperature. The audit is intentionally targeted, not a full cross-species uncertainty quantification; absolute TES values are therefore best read as relative rankings under a documented parameter set.
8. **Body condition index approximation.** We model blubber depletion as increased conductance ($k_{\text{eff}} = k/\text{BCI}$). This is a conductance proxy for emaciation, not a literal representation of simultaneous thickness loss, lipid depletion and tissue-water changes. It also does not resolve the heterogeneous spatial pattern of blubber catabolism, which can proceed faster from the axilla and peduncle regions than from mid-body (Koopman, 2007). Future implementations should apply region-specific BCI modifiers.
9. **Cross-species transfer of the counter-current calibration.** The counter-current terms (appendage retia, peripheral shell) are anchored on the minke, the only species with a stratified temperature transect (Folkow and Blix, 1992), and applied to the other species as conserved physiology (the shell and taper depths scale with local blubber thickness). This transfer is only partially adequate for the most heavily insulated species: because the peripheral shell cools the interface in proportion to its resistance relative to the blubber, the thick blubber of large whales (e.g. 28 cm in the bowhead configuration) leaves the modelled core/blubber interface warmer

than in the thin-blubbered minke. No stratified transect exists for these species to test the prediction, so their interface temperatures should be read as model outputs rather than validated values. They are a clear target for future physiological measurement.

10. **Field metabolic rate and the interpretation of the energy estimates.** The species field metabolic rate used to anchor the metabolic source is a resting economical rate ($\sim 50\%$ of Kleiber for large rorquals; Blawas et al., 2025). The whole-animal heat loss Q_{loss} that the model returns at the defended core temperature is instead the total heat production the animal must sustain. This is a genuine *field* rate, $\sim 2\text{--}3\times$ Kleiber in cold water. Following the heat increment of feeding (Section 2.9), it is fuelled by metabolizable energy intake. We therefore report Q_{loss} -derived energy requirements as maintenance-equivalent heat-loss costs with uncertain absolute scale: they carry the $\sim 1.5\times$ Pennes-reservoir over-prediction quantified against the minke, whereas the *relative* differences between temperatures and species are more robust to this bias. These values are not full daily rations; measured feeding-season intake also supports activity, migration, growth, reproduction and energy storage (Savoca et al., 2021).
11. **Geometry provenance and the scarcity of scientific cetacean 3D models.** A fundamental constraint on multi-species cetacean FEM studies is the near-absence of publicly available, scientifically validated 3D body geometries. Of the seven species modelled here, four are derived from photogrammetric scans or anatomist-validated designs: humpback whale (CARI'MAM project, Weeteam 2020) and the right, blue and gray whales (Digital Life, University of Massachusetts Amherst, Digital Life Project, University of Massachusetts Amherst 2019; Segre et al. 2023; Digital Life Project, University of Massachusetts Amherst 2020). The remaining three are artist CG models. The fin and sei whales come from named, CC-licensed sources (Woelper, 2017; Lvov, 2018). The bowhead uses a dedicated *Balaena mysticetus* model (artist “3D Molier”), which in this revision replaces the right-whale proxy used previously. The surface detail of these three models is unverified against measured morphometrics. This scarcity reflects a broader gap in the literature: unlike terrestrial megafauna, large cetaceans are rarely accessible for controlled photogrammetric scanning, and post-stranding models are morphologically distorted. The thermal outputs for species with unverified geometries should be interpreted as body-plan archetypes rather than species-exact predictions. We encourage future work to prioritise open-access, peer-reviewed 3D cetacean models as foundational data infrastructure for physiological modelling.

5. Conclusions

We have developed a three-dimensional finite element thermal model for baleen whales and applied it to seven mysticete species from three families (Balaenopteridae, Balaenidae, and Eschrichtiidae), using a water-temperature-based thermoregulatory control scheme and food-stress extensions, with a common minke whale as a physiological validation case using proxy geometry. In total, the main comparative/validation matrix contains **793 finite-element simulations**. This includes an identical 99-case grid (nine water temperatures $\times$ four thermoregulatory modes $\times$ a food-availability $\times$ body-condition extension) for each of the eight whale models, plus one minke Folkow and Blix calibration run. The seven comparative species supply the 693 cases behind the comparative results. A targeted 48-case humpback sensitivity audit was then added for the weakest-constrained thermoregulatory parameters. The principal findings are:

1. Mysticetes employ two distinct but complementary thermoregulatory strategies, passive insulation (thick blubber) and active vascular control (thermal windows), with the relative importance varying predictably with family-level body plan. Active vascular control modulates total heat loss by up to 53% in rorquals without substantially altering core temperature; its primary function is energetic, not thermoprotective.
2. Under the remediated counter-current physiology, thermal windows (pectoral flippers, flukes, ventral grooves, dorsal fin) account for a substantial and comparable fraction of total heat dissipation across all seven species (42–62% at 10 °C). A well-insulated trunk makes the relatively bare appendages the path of least resistance even in thick-blubbered balaenids. The family-level distinction between insulation and active-window strategies is therefore carried by the absolute and mass-specific magnitude of heat loss, not by the appendage fraction.
3. The thermal-maintenance energy requirement is quantified for every species and water temperature. On a mass-specific basis the heavily blubbered balaenids (bowhead, right whale) require roughly one third the energy per unit body mass of the thermal-window rorquals, and for most species the requirement peaks in the mid-cold range where active window closure has saturated. These magnitudes are maintenance-equivalent heat-loss costs; absolute values should be read cautiously, with relative contrasts carrying the primary inference (Section 4.5).

4. Under the defended arterial-set-point formulation, no simulation produces acute core-temperature failure: no case in the corrected matrix crosses a physiological limit. The thermal diagnostic is therefore the elevated heat production, and thus the maintenance-equivalent energy demand, required to hold the defended state. We quantify this per species and temperature.
5. The Thermoregulatory Effort Score (TES) provides a species-and-temperature specific index of cardiovascular demand. Across the ecological range every species operates in the conserving half of its perfusion capacity, from strong vasoconstriction (humpback TES ≈ -0.90 at $T_{\infty} \leq 10^{\circ}\text{C}$) to only mild vasoconstriction near baseline perfusion in the warmest water (bowhead TES ≈ -0.37); no species reaches sustained active vasodilation, underscoring that heat conservation, not heat dumping, is the operative demand in these large cold-water endotherms.
6. Swimming provides no thermoregulatory subsidy: sustained locomotion raises the defended core temperature only modestly ($+0.06$ to $+0.31^{\circ}\text{C}$) but increases total surface heat loss and thermal-maintenance cost at every water temperature ($+8$ to $+15\%$ for the humpback), so cold-water migration adds to rather than offsets the maintenance requirement.
7. The thermal sensitivity index ($S = 260\text{ W }^{\circ}\text{C}^{-1}$ to $1458\text{ W }^{\circ}\text{C}^{-1}$, from gray whale to blue whale, scaling primarily with body size) and the smooth, sometimes mid-cold-peaked $\dot{Q}$ profiles produced by the tanh control signal provide a quantitative ranking of species sensitivity to ocean temperature change.
8. The 48-case humpback sensitivity audit supports the qualitative robustness of the central result: across tested ranges for T_{tn} , λ , CCHE effectiveness, appendage perfusion, h , blubber conductance and BCI, all cases survived and the largest core-temperature shift was only 0.12°C . Absolute maintenance-equivalent energy costs varied by up to 22% , so those magnitudes should be interpreted with the uncertainty cautions stated above.

The biomarker framework introduced here includes TES, thermal efficiency, thermal window fraction, and the thermal-maintenance energy requirement. It provides quantitative inputs for bioenergetic population models. These are particularly relevant for conservation of North Atlantic right whales, where a substantial cold-water heat-production requirement, limited heat-dissipation anatomy, and documented prey decline create compounding physiological stressors that act through the animal's energy budget rather than through acute thermal failure. Future extensions will incorporate facultative shivering thermogenesis, an explicit Weinbaum and Jiji treatment of the deep circulation to tighten the absolute heat-loss magnitudes, region-specific body-condition profiles, and ontogenetic scaling to assess thermal vulnerability in calves and juveniles.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code, species configuration files, mesh-generation scripts, and post-processing scripts are available from the corresponding author upon reasonable request. The manuscript-level output summaries are available on request; large FEM field files are retained locally because of storage size.

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CRedit authorship contribution statement

Domagoj K. Hackenberger: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing: Original Draft, Writing: Review and Editing, Visualization. **Tamara Djerdj:** Formal analysis, Investigation, Validation, Writing: Review and Editing. **Branimir K. Hackenberger:** Supervision, Formal analysis, Writing: Review and Editing.

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A. Glossary

A.1. Abbreviations

Term	Definition
BCI	Body-condition index; dimensionless multiplier describing the retained blubber/condition state in food-stress scenarios.
CCHE	Counter-current heat exchange; vascular heat recovery mechanism that reduces net heat loss through appendages and peripheral tissues.
FA	Food availability; dimensionless fraction of baseline food intake used to scale metabolic heat input in food-stress simulations.
$FA_{\min}$	Minimum food availability required for metabolic heat balance under the model assumptions. Values clamped at 1.0 indicate that balance is not achievable through food intake alone.
FEM	Finite element method; numerical discretisation used to solve the three-dimensional bioheat equation.
FMR	Field metabolic rate; whole-animal metabolic target used to scale the volumetric metabolic source.
HIF	Heat increment of feeding; post-ingestive heat production associated with digestion and assimilation.
TES	Thermoregulatory Effort Score; signed index from -1 (maximum vasoconstriction) to $+1$ (maximum vasodilation), with 0 denoting thermoneutral baseline.
TWF	Thermal-window fraction; fraction of total surface heat loss passing through appendage and other thermal-window regions.

A.2. Symbols

Symbol	Definition
T_{∞}, T_w	Ambient water temperature imposed at the skin-to-water boundary.
T_{core}	Species core/arterial temperature set-point or reported deep-core temperature, depending on context.
T_{tn}	Thermoneutral water temperature used in the active-control law.
T_s	Skin or external surface temperature.
T_a	Arterial blood temperature in the Pennes perfusion term.
u	Water-temperature control signal, $u = \tanh[(T_{\infty} - T_{\text{tn}})/\lambda]$.
λ	Bandwidth of the active-control signal.
$\dot{Q}_{\text{loss}}$	Total steady-state surface heat loss.
$\dot{Q}_{\text{met}}$	Total metabolic heat production.
$\dot{Q}_{\text{core}}, \dot{Q}_{\text{musc}}$	Pre-scaling volumetric metabolic heat generation assigned to core and muscle tissue.
η	Heat-balance efficiency, $\eta = \dot{Q}_{\text{met}}/\dot{Q}_{\text{loss}}$.
R_{eff}	Effective whole-body thermal resistance, $(T_{\text{max}} - T_{\infty})/\dot{Q}_{\text{loss}}$.
S	Thermal sensitivity index, change in heat loss per degree of water temperature change.
h	Effective surface convection coefficient.
k	Tissue thermal conductivity.
ω_b	Volumetric blood perfusion rate.
c_b	Specific heat capacity of blood.
s_r	Dimensionless effective appendage-window perfusion implied by CCHE effectiveness, $s_r = 1 - \varepsilon_r$.
ϕ_r	Region-specific vascular/perfusion scaling factor retained for legacy configuration fields and output metadata.
$\varepsilon_{\text{rete}}$	Counter-current heat-exchange effectiveness of the appendage-root retia.

B. Species configuration parameters

The complete set of species-specific parameters used in the simulations is provided in the supplementary material as machine-readable YAML files. Table 20 summarises the key parameters.

Table 20: Summary of species-specific simulation parameters.

Species	T_{core} (°C)	T_{tn} (°C)	k_{blub} (W m ⁻¹ K ⁻¹)	f_{perf}	Q_{core} (W m ⁻³)	Q_{musc} (W m ⁻³)
Humpback	36.8	25	0.21	0.012	3000	700
Right whale	36.6	8	0.20	0.012	3200	750
Blue whale	36.7	10	0.22	0.012	2800	650
Bowhead	33.8	0	0.19	0.012	3200	750
Fin whale	36.1	8	0.22	0.012	2900	680
Gray whale	36.0	8	0.20	0.012	3000	700
Sei whale	36.1	10	0.22	0.012	3000	700
Minke [†]	35.0	10	0.25	0.012	3000	700

[†]Minke validation species (Section 4.4); arterial set-point 35.0 °C (deep body 34.7 °C, Folkow and Blix, 1992), measured blubber conductivity 0.25 W m⁻¹ K⁻¹ (Kvadsheim et al., 1996). f_{perf} : perfusion scaling factor (CCHE reduction); Q_{core} , Q_{musc} : pre-scaling core and muscle metabolic heat. The effective metabolic activity fraction f_{vol} is not a model input but a derived output of the FMR-forward scaling (Section 2.4.4), reported per run in the output metadata.

C. Mesh convergence study

To verify mesh independence, we performed a convergence study on the humpback whale geometry using four successively refined meshes (Table 21). The reference solution was obtained on the finest mesh. At the production mesh resolution (0.15 m element size, ~176 000 nodes), the maximum temperature error relative to the finest mesh was less than 0.1 °C, and the total heat loss error was less than 1%.

Table 21: Mesh convergence study for the humpback whale at $T_{\infty} = 10^{\circ}\text{C}$.

Element size (m)	Nodes	Elements	$T_{\text{core}} (^{\circ}\text{C})$	$\Delta\dot{Q} (\%)$
0.30	28 000	150 000	36.62	3.2
0.20	78 000	430 000	36.71	1.1
0.15	176 348	982 853	36.74	0.4
0.10	520 000	3 100 000	36.76	—